Set 73: Spotting image flaws and garbled text

Outcome AI.IMAGE.02: Inspect a generated image for visual errors and unreadable or misspelled text

Module: Creative: images, design and presentations

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
Generated images often contain FLAWS that are easy to miss at a glance. Look closely at HANDS and FINGERS (extra, missing or bent oddly), EYES and TEETH, OBJECTS that melt into each other, SHADOWS and REFLECTIONS that do not match the light, and PATTERNS that break, such as a fence that changes shape. TEXT inside images is a common problem: letters may be garbled, words misspelled ("Bake Sael"), or numbers changed. Before using an image, zoom in, read every word letter by letter, and fix or remove what is wrong. Often it is easier to add text yourself in a design app.

Which part of a generated image often has errors?

A. The image size

B. The overall color of the sky, which image tools always get wrong in every picture

C. The file name

D. Hands and fingers

REFERENCE LABEL: NOT PART OF THE QUESTION ID S73-M-Q1
Artificial Intelligence, Set 73: Spotting image flaws and garbled text · MAIN · Q1/5 · Outcome AI.IMAGE.02 (local)

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Prepared example (a written description of a fictional generated image)

Image description: a cheerful flyer with cupcakes. Visible text reads "SPRING BAKE SAEL - Saturday, April 31 - 10 am to 2 pm - Room 12."

In this prepared example, which word is misspelled?

A. Room

B. Saturday

C. SAEL

D. SPRING

REFERENCE LABEL: NOT PART OF THE QUESTION ID S73-M-Q2

Artificial Intelligence, Set 73: Spotting image flaws and garbled text · MAIN · Q2/5 · Outcome AI.IMAGE.02 (local)

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Prepared example (a written description of a fictional generated image)

Image description: a cheerful flyer with cupcakes. Visible text reads "SPRING BAKE SAEL - Saturday, April 31 - 10 am to 2 pm - Room 12."

In the same example, what other error appears?

A. Room 12 is not a real number

B. April 31 does not exist

C. Saturday is misspelled

D. 10 am to 2 pm is too long

REFERENCE LABEL: NOT PART OF THE QUESTION ID S73-M-Q3

Artificial Intelligence, Set 73: Spotting image flaws and garbled text · MAIN · Q3/5 · Outcome AI.IMAGE.02 (local)

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How should you check text in an image?

A. Zoom in and read it letter by letter
B. Trust it if the image is pretty
C. Count the colors
D. Glance at it quickly from a distance

REFERENCE LABEL: NOT PART OF THE QUESTION ID S73-M-Q4
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What is often easier than fixing text the generator made?

A. Leaving the errors in
B. Adding the text yourself in a design app
C. Removing all images
D. Asking for the same image again and again

REFERENCE LABEL: NOT PART OF THE QUESTION ID S73-M-Q5
Artificial Intelligence, Set 73: Spotting image flaws and garbled text · MAIN · Q5/5 · Outcome AI.IMAGE.02 (local)

Learn
Use a five-point checklist. (1) BODIES: count fingers, check eyes, teeth and limbs. (2) OBJECTS: look for things blending together or floating. (3) LIGHT: do shadows fall the same way? (4) PATTERNS: do fences, tiles and stripes stay consistent? (5) TEXT: read every word letter by letter, and check dates and numbers. Example: a picture shows a person waving with six fingers, a cup merging into a table, and a sign reading "Opne." That fails points 1, 2 and 5.

Six fingers on a waving hand fails which point?

A. Text
B. Patterns
C. Light
D. Bodies

REFERENCE LABEL: NOT PART OF THE QUESTION ID S73-B1-Q1
Artificial Intelligence, Set 73: Spotting image flaws and garbled text · BACKWARD 1 (underlying skill) · Q1/5 · Outcome AI.IMAGE.02 (local)

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A cup merging into a table fails which point?

A. Objects
B. Text
C. Light
D. Bodies

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A sign reading "Opne" fails which point?

A. Objects
B. Bodies
C. Text
D. Patterns

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Shadows falling in two different directions fail which point?

A. Text
B. Bodies
C. Objects
D. Light

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A striped shirt whose stripes change direction fails which point?

A. Bodies
B. Light
C. Patterns
D. Text

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A poster image shows the event phone number with one digit changed. What should you do?

A. Remove the event date too
B. Fix the number before printing
C. Make the number smaller
D. Print it anyway

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A family portrait for a slideshow has a child with three arms. What is the best choice?

A. Regenerate, edit or choose another image

B. Make it black and white

C. Keep it, since viewers rarely look closely at people in pictures shown during slideshows

D. Add a caption explaining it

REFERENCE LABEL: NOT PART OF THE QUESTION ID S73-F1-Q2
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A store sign in a generated image reads "Grocey Stroe." What kind of flaw is this?

A. No flaw
B. Garbled text
C. A pattern flaw
D. A lighting flaw

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Before posting a generated image online, what is the best first step?

A. Post it right away, since generated images are checked for errors before you receive them

B. Add a filter

C. Zoom in and check details

D. Compress it

REFERENCE LABEL: NOT PART OF THE QUESTION ID S73-F1-Q4
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A flyer has correct text that you added yourself and a generated background. What still needs checking?

A. Only the font

B. Nothing, because adding text yourself also fixes every flaw in the generated background

C. Only the file size

D. The background for visual flaws

REFERENCE LABEL: NOT PART OF THE QUESTION ID S73-F1-Q5
Artificial Intelligence, Set 73: Spotting image flaws and garbled text · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.IMAGE.02 (local)

Learn
Some flaws matter more than others. A slightly odd cloud is harmless; a wrong date or price on a flyer is not. A misspelled name on a certificate or a wrong phone number on a poster can cause real problems. Small symbols, such as safety signs or logos, may look almost right but be wrong or belong to someone else. Check where the image will be seen: details that look fine on a phone can be obvious when printed large.

Which flaw matters most on an event flyer?

A. A wrong date
B. A tree with odd leaves
C. A slightly blurry corner
D. A cloud in the background

REFERENCE LABEL: NOT PART OF THE QUESTION ID S73-F2-Q1
Artificial Intelligence, Set 73: Spotting image flaws and garbled text · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.IMAGE.02 (local)

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A certificate image misspells the winner's name. Why is this serious?

A. It changes the paper size

B. It is not serious, because names on certificates are decorative and nobody reads them

C. It makes the file larger

D. It gives wrong information about a real person

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A generated safety sign looks almost like a real one but the symbol is slightly different. What is the risk?

A. The colors fade

B. People may misread the warning

C. No risk, because people always understand safety signs even when the symbol is changed

D. The sign becomes funny

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An image looks fine on a phone. Why check before printing a large banner?

A. Flaws can be obvious when enlarged

B. Because banners never show text

C. Because printing always changes every color in a picture into completely different colors

D. Because phones show extra details

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A generated logo looks like another company's logo. What should you do?

A. Add your name and use it
B. Change one color and use it
C. Avoid using it
D. Use it

REFERENCE LABEL: NOT PART OF THE QUESTION ID S73-F2-Q5
Artificial Intelligence, Set 73: Spotting image flaws and garbled text · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.IMAGE.02 (local)

Set 74: Consistency and editing

Outcome AI.IMAGE.03: Keep characters, colors and details consistent across a series, and request targeted edits

Module: Creative: images, design and presentations

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When you need a SERIES of images, such as a character on every page of a story or slides with one look, generators often drift: hair color changes, a shirt gets new stripes, the art style shifts. To keep CONSISTENCY, reuse the same detailed description each time ("Mia: short curly black hair, yellow raincoat, round glasses"), keep the same style words, and use reference-image features if your tool has them. For EDITS, ask for one TARGETED change ("make only the raincoat green; keep everything else") instead of redescribing the whole image, and compare the result with the original.

What is drift in an image series?

A. Images loading slowly
B. Images getting larger
C. Details changing from one image to the next
D. Images slowly moving across the screen

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How can you keep a character consistent?

A. Reuse the same detailed description each time
B. Change the style each page
C. Leave out the description
D. Describe the character differently each time

REFERENCE LABEL: NOT PART OF THE QUESTION ID S74-M-Q2
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Which edit request is targeted?

A. "Make it different."
B. "Change it."
C. "Redraw the whole picture again from the beginning."
D. "Make only the raincoat green; keep everything else."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S74-M-Q3
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After an edit, what should you do?

A. Post it immediately
B. Compare the result with the original
C. Make five more edits without looking
D. Delete the original image right away

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What tool feature can help consistency?

A. Reference images
B. A faster internet connection
C. A larger screen
D. A feature that makes every image

REFERENCE LABEL: NOT PART OF THE QUESTION ID S74-M-Q5
Artificial Intelligence, Set 74: Consistency and editing · MAIN · Q5/5 · Outcome AI.IMAGE.03 (local)

Learn
A school uniform keeps students looking alike: same colors, same shirt, same logo every day. If each student chose their own outfit daily, photos from the year would look mismatched. A character in a series of images needs a "uniform" too: the same written description of hair, clothes and colors, used every time, so the character looks like the same person on every page.

What does a uniform keep the same?

A. The timetable
B. The weather
C. Each student's height
D. Colors, clothes and logo

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Artificial Intelligence, Set 74: Consistency and editing · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.IMAGE.03 (local)

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Students choosing new outfits daily is like what?

A. A targeted edit
B. A reference image
C. Drift in an image series
D. A perfectly consistent series

REFERENCE LABEL: NOT PART OF THE QUESTION ID S74-B1-Q2
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A character's "uniform" is what?

A. A new style each page
B. The same written description used every time
C. No description
D. A different description for every page

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Why do mismatched photos look odd?

A. Things that should match do not
B. Because uniforms are blue
C. Because photos taken on different days always come out blurry and unusable
D. Because schools are large

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What is the lesson for an image series?

A. Use only one image
B. Avoid characters
C. Change everything each time
D. Keep the same details each time

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A children's story has a dog that is brown on page 1 and gray on page 3. What should the author do?

A. Use gray on every page without saying so
B. Remove the dog
C. Reuse one exact description of the dog for every page
D. Change the story so that the dog changes color by magic

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Nia wants to change only the background of an image from a park to a beach. Which request fits?

A. "Beach."

B. "Change only the background to a beach; keep the people the same."

C. "Change everything to look like summer."

D. "Make a new beach picture with some people in it who look somewhat like the old ones."

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Artificial Intelligence, Set 74: Consistency and editing · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.IMAGE.03 (local)

Learn
When you need a SERIES of images, such as a character on every page of a story or slides with one look, generators often drift: hair color changes, a shirt gets new stripes, the art style shifts. To keep CONSISTENCY, reuse the same detailed description each time ("Mia: short curly black hair, yellow raincoat, round glasses"), keep the same style words, and use reference-image features if your tool has them. For EDITS, ask for one TARGETED change ("make only the raincoat green; keep everything else") instead of redescribing the whole image, and compare the result with the original.

A slide series uses watercolor on some slides and photos on others. What should change?

A. Use the same style words on every slide

B. Use random styles

C. Add more styles, so the slides look varied and keep the audience's attention throughout

D. Remove all images

REFERENCE LABEL: NOT PART OF THE QUESTION ID S74-F1-Q3
Artificial Intelligence, Set 74: Consistency and editing · FORWARD 1: APPLY IT · Q3/5 · Outcome AI.IMAGE.03 (local)

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After asking to change a shirt to blue, what should you check?

A. Only the file name

B. Nothing, because a targeted edit request always changes exactly one thing in any picture

C. That only the shirt changed

D. Only the size

REFERENCE LABEL: NOT PART OF THE QUESTION ID S74-F1-Q4
Artificial Intelligence, Set 74: Consistency and editing · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.IMAGE.03 (local)

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Which description helps a character stay consistent?

A. "A boy."

B. "Leo: tall, red cap, green scarf, freckles."

C. "Leo."

D. "Leo: a boy, and he looks however you think is best for the story on each page."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S74-F1-Q5
Artificial Intelligence, Set 74: Consistency and editing · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.IMAGE.03 (local)

Learn
Edits can change more than you asked: a request to change the background may also change a face or move objects. Check the whole image after each edit, not just the part you changed. Keep a copy of each version so you can go back. For a long series, write a short "style sheet" listing each character's features, the colors and the style, and check each new image against it before accepting it.

You asked to change the sky, and the person's face also changed. What does this show?

A. That the sky and face are the same
B. That edits never work
C. That the tool always changes faces
D. Edits can change more than asked

REFERENCE LABEL: NOT PART OF THE QUESTION ID S74-F2-Q1
Artificial Intelligence, Set 74: Consistency and editing · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.IMAGE.03 (local)

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Why keep a copy of each version during edits?

A. So you can go back if an edit goes wrong
B. So images look brighter
C. To use more storage
D. Because tools refuse to edit any image

REFERENCE LABEL: NOT PART OF THE QUESTION ID S74-F2-Q2
Artificial Intelligence, Set 74: Consistency and editing · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.IMAGE.03 (local)

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What is a style sheet for an image series?

A. A list of fonts only
B. A list of each character's features, colors and style
C. A spreadsheet of prices
D. A document that lists the printer settings for every page

REFERENCE LABEL: NOT PART OF THE QUESTION ID S74-F2-Q3
Artificial Intelligence, Set 74: Consistency and editing · FORWARD 2: HANDLE THE COMPLICATIONS · Q3/5 · Outcome AI.IMAGE.03 (local)

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A new image almost matches, but the character's glasses are missing. What should you do?

A. Remove glasses from all other images
B. Change the character's name
C. Reject or fix it using the style sheet
D. Accept it

REFERENCE LABEL: NOT PART OF THE QUESTION ID S74-F2-Q4
Artificial Intelligence, Set 74: Consistency and editing · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.IMAGE.03 (local)

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When should you check a new image against the style sheet?

A. After the whole series is printed and handed out, so you can fix it next time
B. Only for the first image
C. Never
D. Before accepting it

REFERENCE LABEL: NOT PART OF THE QUESTION ID S74-F2-Q5
Artificial Intelligence, Set 74: Consistency and editing · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.IMAGE.03 (local)

Set 75: Diagrams and factual accuracy

Outcome AI.IMAGE.04: Check generated diagrams, maps and charts against real data before using them

Module: Creative: images, design and presentations

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
Image generators make pictures that LOOK like diagrams, maps and charts, but they do not check facts or measure data. A generated map may place cities in the wrong spots or invent roads; a chart may have bars whose heights do not match their labels; a science diagram may label the wrong part. For anything meant to INFORM, build the diagram from real data or a trusted source, for example with a spreadsheet chart or an official map, or check every label, number and position against that source. Use generated images for decoration, not as evidence.

Do image generators check facts when they draw a map?

A. No; they make pictures that look like maps

B. Yes; every generated map is checked against official satellite data before it is shown

C. Only for small countries

D. Only in color

REFERENCE LABEL: NOT PART OF THE QUESTION ID S75-M-Q1
Artificial Intelligence, Set 75: Diagrams and factual accuracy · MAIN · Q1/5 · Outcome AI.IMAGE.04 (local)

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Prepared example (a written description of a fictional generated bar chart)

Bar label	Value printed on bar	Bar height drawn (grid squares)
Monday	10	1
Tuesday	40	4
Wednesday	20	5

In this prepared example, which bar is drawn wrongly?

A. Tuesday

B. Wednesday

C. None

D. Monday

REFERENCE LABEL: NOT PART OF THE QUESTION ID S75-M-Q2

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How should you make a chart meant to inform?

A. Copy a chart from memory

B. Guess the numbers

C. Build it from real data, such as in a spreadsheet

D. Ask an image generator to draw a chart that looks professional and has lots of colors

REFERENCE LABEL: NOT PART OF THE QUESTION ID S75-M-Q3
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What should generated images be used for?

A. Medical diagrams for patients
B. Official maps
C. Evidence in reports
D. Decoration, not evidence

REFERENCE LABEL: NOT PART OF THE QUESTION ID S75-M-Q4
Artificial Intelligence, Set 75: Diagrams and factual accuracy · MAIN · Q4/5 · Outcome AI.IMAGE.04 (local)

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A generated science diagram labels the heart as the lungs. What does this show?

A. That the tool is always right

B. Labels can be wrong

C. That diagrams cannot have labels

D. That the heart and lungs are the same organ in some science textbooks used in schools

REFERENCE LABEL: NOT PART OF THE QUESTION ID S75-M-Q5
Artificial Intelligence, Set 75: Diagrams and factual accuracy · MAIN · Q5/5 · Outcome AI.IMAGE.04 (local)

Learn
To check a bar chart, compare each bar's HEIGHT with its NUMBER. The biggest number should have the tallest bar, the smallest number the shortest, and a number twice as large should look about twice as tall. Also read the AXIS: does it start at 0? Example: bars labeled A = 5, B = 10, C = 15. C should be tallest, A shortest, and B about twice A. If B is drawn shorter than A, the chart is wrong.

A = 5, B = 10, C = 15. Which bar should be tallest?

A. B

B. A, because the first bar in a chart is always drawn as the tallest one

C. C

D. All equal

REFERENCE LABEL: NOT PART OF THE QUESTION ID S75-B1-Q1
Artificial Intelligence, Set 75: Diagrams and factual accuracy · BACKWARD 1 (underlying skill) · Q1/5 · Outcome AI.IMAGE.04 (local)

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Which should be shortest?

A. A

B. B

C. C, since the last bar in any chart is usually drawn the shortest of all

D. None

REFERENCE LABEL: NOT PART OF THE QUESTION ID S75-B1-Q2
Artificial Intelligence, Set 75: Diagrams and factual accuracy · BACKWARD 1 (underlying skill) · Q2/5 · Outcome AI.IMAGE.04 (local)

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B = 10 and A = 5. How should B compare with A?

A. Half as tall
B. Ten times taller
C. About the same height
D. About twice as tall

REFERENCE LABEL: NOT PART OF THE QUESTION ID S75-B1-Q3
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B is drawn shorter than A. What does that mean?

A. The chart is wrong
B. A must be 20
C. B must be 3
D. The chart is correct

REFERENCE LABEL: NOT PART OF THE QUESTION ID S75-B1-Q4
Artificial Intelligence, Set 75: Diagrams and factual accuracy · BACKWARD 1 (underlying skill) · Q4/5 · Outcome AI.IMAGE.04 (local)

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What else should you read?

A. The chart's font

B. The file name

C. Whether the axis starts at 0

D. The color of each bar, because colors show the real values more clearly than heights

REFERENCE LABEL: NOT PART OF THE QUESTION ID S75-B1-Q5
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Ana needs a map of bus routes for new students. What is the best source?

A. An image generator asked to draw a detailed and accurate map of the city's bus routes

B. The transit agency's official map

C. A map of another city

D. A sketch from memory

REFERENCE LABEL: NOT PART OF THE QUESTION ID S75-F1-Q1
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A generated chart for a club report shows membership numbers. What should be checked?

A. Nothing

B. Only that the colors match the club logo, since the numbers come from the tool and are fine

C. Only the title

D. Every number against club records

REFERENCE LABEL: NOT PART OF THE QUESTION ID S75-F1-Q2
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A teacher wants a pretty background picture of planets for a slide title. Is a generated image reasonable?

A. No, because generated images can never be used for anything at all in any classroom setting

B. Yes, as decoration labeled as an illustration

C. Only if it shows text

D. Yes, and it can be used as a scientific diagram too

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Which is the most accurate way to make a pie chart of a survey?

A. Describe the survey to an image tool and ask for a pie chart that looks about right

B. Use a chart from another survey

C. Enter the survey results into a spreadsheet chart

D. Draw slices by guessing

REFERENCE LABEL: NOT PART OF THE QUESTION ID S75-F1-Q4
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A generated diagram of a plant has a part labeled "photosynthesis root." What should you do?

A. Check labels against a trusted source

B. Add more labels

C. Make it bigger

D. Keep it, since diagrams always use correct scientific names even when they look unusual

REFERENCE LABEL: NOT PART OF THE QUESTION ID S75-F1-Q5
Artificial Intelligence, Set 75: Diagrams and factual accuracy · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.IMAGE.04 (local)

Learn
Some errors are subtle. A chart's axis may start at 50 instead of 0, making small differences look huge. A diagram may be mostly right but swap two labels. A map may have the right shape but the wrong scale. Read axes and legends, count items, and compare with the source. If you use a generated picture as an illustration next to real data, label it clearly as an illustration so readers do not treat it as accurate.

A chart's axis starts at 50, so 52 looks twice as tall as 51. What is the problem?

A. The numbers are wrong
B. The chart is too colorful
C. No problem
D. Small differences look huge

REFERENCE LABEL: NOT PART OF THE QUESTION ID S75-F2-Q1
Artificial Intelligence, Set 75: Diagrams and factual accuracy · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.IMAGE.04 (local)

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A diagram is right except two labels are swapped. Why is it still a problem?

A. Because diagrams need three labels
B. Because swapped labels change the colors
C. Readers learn the wrong names
D. It is not a problem

REFERENCE LABEL: NOT PART OF THE QUESTION ID S75-F2-Q2
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A map has the right shape but the wrong scale. What could go wrong?

A. Distances would be misjudged
B. Street names would change
C. The colors would fade
D. Nothing

REFERENCE LABEL: NOT PART OF THE QUESTION ID S75-F2-Q3
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Why label a generated picture as "illustration" next to real data?

A. Because labels are required by every printer
B. So readers do not treat it as accurate
C. To make the file smaller
D. So readers know the illustration is more accurate

REFERENCE LABEL: NOT PART OF THE QUESTION ID S75-F2-Q4
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Which habit catches most chart errors?

A. Checking the font
B. Counting the colors
C. Looking at the overall shape of the chart for a second
D. Reading axes and legends and comparing with the source

REFERENCE LABEL: NOT PART OF THE QUESTION ID S75-F2-Q5
Artificial Intelligence, Set 75: Diagrams and factual accuracy · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.IMAGE.04 (local)

Set 76: Presentations: structure, visuals and export

Outcome AI.IMAGE.05: Plan a clear presentation and tell a text draft from the separate step of creating and exporting a file

Module: Creative: images, design and presentations

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Learn
A good PRESENTATION has a clear STRUCTURE: a purpose, a few main points in order, evidence for each, and a closing message, usually with one idea per slide and few words. AI tools can help OUTLINE slides and suggest visuals. Keep two steps apart. (1) A TEXT DRAFT: the tool writes slide titles, bullet points and speaker notes as text. (2) A FILE: an actual slide file must be created in a presentation app or by a tool feature that really makes files. Many chats only produce text, even when they describe "slides." After any file is created or exported, open it and check every slide.

What does a clear presentation structure include?

A. A purpose, main points with evidence, and a close

B. Only pictures

C. Only a title slide

D. As many slides as possible, each packed with full paragraphs of text so nothing is left out

REFERENCE LABEL: NOT PART OF THE QUESTION ID S76-M-Q1
Artificial Intelligence, Set 76: Presentations: structure, visuals and export · MAIN · Q1/5 · Outcome AI.IMAGE.05 (local)

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Fictional example (prepared in advance; not a real AI output)

Request:

Make me a 5-slide presentation on recycling.

Response:

Slide 1: Why Recycling Matters
Slide 2: What Can Be Recycled
Slide 3: How Our School Recycles
Slide 4: Common Mistakes
Slide 5: What You Can Do
(Each slide has 3 bullet points and speaker notes below.)

In this fictional example, what did the tool produce?

A. A finished slide file saved on the computer that is ready to open in any presentation app

B. A video

C. A text outline of slides

D. A printed handout

REFERENCE LABEL: NOT PART OF THE QUESTION ID S76-M-Q2
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How is an actual slide file usually made?

A. By typing the word "slides" into a chat, which always creates a real file automatically

B. By printing the chat

C. By renaming a text file

D. In a presentation app or with a real file-making feature

REFERENCE LABEL: NOT PART OF THE QUESTION ID S76-M-Q3
Artificial Intelligence, Set 76: Presentations: structure, visuals and export · MAIN · Q3/5 · Outcome AI.IMAGE.05 (local)

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What is a common rule for slides?

A. Use full paragraphs
B. One idea per slide, few words
C. No titles
D. Put every idea from the whole talk

REFERENCE LABEL: NOT PART OF THE QUESTION ID S76-M-Q4
Artificial Intelligence, Set 76: Presentations: structure, visuals and export · MAIN · Q4/5 · Outcome AI.IMAGE.05 (local)

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After a slide file is created, what should you do?

A. Delete the outline
B. Email it unopened
C. Present it straight away
D. Open it and check every slide

REFERENCE LABEL: NOT PART OF THE QUESTION ID S76-M-Q5
Artificial Intelligence, Set 76: Presentations: structure, visuals and export · MAIN · Q5/5 · Outcome AI.IMAGE.05 (local)

Learn
A RECIPE tells you the dishes, ingredients and steps, but you cannot eat it. The MEAL exists only after someone cooks it in a kitchen. A text draft of slides is like a recipe: it lists titles, points and notes. The slide FILE is like the cooked meal: it has to be made in a presentation app or by a feature that really creates files. And just as you taste a meal before serving it, you open and check the file before presenting.

A text draft of slides is like what?

A. A grocery store
B. A recipe
C. A kitchen
D. A cooked meal

REFERENCE LABEL: NOT PART OF THE QUESTION ID S76-B1-Q1
Artificial Intelligence, Set 76: Presentations: structure, visuals and export · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.IMAGE.05 (local)

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The slide file is like what?

A. The cooked meal
B. The oven
C. The shopping list
D. The recipe card

REFERENCE LABEL: NOT PART OF THE QUESTION ID S76-B1-Q2
Artificial Intelligence, Set 76: Presentations: structure, visuals and export · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.IMAGE.05 (local)

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Where is the meal made?

A. At the table
B. Inside the recipe book itself, as soon as someone reads the recipe out loud
C. In a kitchen
D. In the store

REFERENCE LABEL: NOT PART OF THE QUESTION ID S76-B1-Q3
Artificial Intelligence, Set 76: Presentations: structure, visuals and export · BACKWARD 1 (concept simplification) · Q3/5 · Outcome AI.IMAGE.05 (local)

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Tasting the meal before serving is like what?

A. Washing dishes
B. Opening and checking the file
C. Buying more food
D. Reading the recipe one more time

REFERENCE LABEL: NOT PART OF THE QUESTION ID S76-B1-Q4
Artificial Intelligence, Set 76: Presentations: structure, visuals and export · BACKWARD 1 (concept simplification) · Q4/5 · Outcome AI.IMAGE.05 (local)

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Can you present a recipe as if it were a meal?

A. Only on holidays

B. Yes, if it is long

C. No; it must be made first

D. Yes, because a recipe and a meal are exactly the same thing when guests arrive

REFERENCE LABEL: NOT PART OF THE QUESTION ID S76-B1-Q5
Artificial Intelligence, Set 76: Presentations: structure, visuals and export · BACKWARD 1 (concept simplification) · Q5/5 · Outcome AI.IMAGE.05 (local)

Learn
A good PRESENTATION has a clear STRUCTURE: a purpose, a few main points in order, evidence for each, and a closing message, usually with one idea per slide and few words. AI tools can help OUTLINE slides and suggest visuals. Keep two steps apart. (1) A TEXT DRAFT: the tool writes slide titles, bullet points and speaker notes as text. (2) A FILE: an actual slide file must be created in a presentation app or by a tool feature that really makes files. Many chats only produce text, even when they describe "slides." After any file is created or exported, open it and check every slide.

Sam asks a chat for "slides" and gets bullet points in the chat window. What should he do next?

A. Present the chat window
B. Give up
C. Look for the slide file in his downloads folder
D. Build the slides in a presentation app from the text

REFERENCE LABEL: NOT PART OF THE QUESTION ID S76-F1-Q1
Artificial Intelligence, Set 76: Presentations: structure, visuals and export · FORWARD 1: APPLY IT · Q1/5 · Outcome AI.IMAGE.05 (local)

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A 10-minute talk has a 30-slide draft. What should change?

A. Cut it to about 5 to 10 slides
B. Remove the closing slide only
C. Add 10 more slides
D. Keep all 30 slides

REFERENCE LABEL: NOT PART OF THE QUESTION ID S76-F1-Q2
Artificial Intelligence, Set 76: Presentations: structure, visuals and export · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.IMAGE.05 (local)

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Which slide follows the "few words" rule?

A. A slide with 12 bullet points

B. A slide with only tiny text

C. A slide with four full paragraphs explaining every detail of recycling from start to finish

D. Title "Recycling Saves Energy" with one chart

REFERENCE LABEL: NOT PART OF THE QUESTION ID S76-F1-Q3
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A tool with a real file feature makes a slide file. What should you check?

A. Only the first slide, because if the first one is fine then every other slide is fine too

B. Nothing

C. Every slide in the opened file

D. Only the file name

REFERENCE LABEL: NOT PART OF THE QUESTION ID S76-F1-Q4
Artificial Intelligence, Set 76: Presentations: structure, visuals and export · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.IMAGE.05 (local)

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Which use of AI fits presentation planning best?

A. Drafting an outline you then revise
B. Skipping preparation
C. Reading the AI text word for word
D. Letting the tool decide the purpose

REFERENCE LABEL: NOT PART OF THE QUESTION ID S76-F1-Q5
Artificial Intelligence, Set 76: Presentations: structure, visuals and export · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.IMAGE.05 (local)

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Exported slides need checking on the device that will show them: fonts may be replaced, images may be missing, animations may not play, and text may overflow boxes. Generated speaker notes may include claims you cannot support, so check them like any other answer. And a good presentation is yours: reorder, cut and rewrite the draft so it fits your audience and the time you have, often about one slide per minute or two.

Slides look fine on a laptop but text overflows on the classroom screen. What should have happened?

A. Nothing, since every screen shows slides in exactly the same way regardless of the computer

B. Checking on the device that will show them

C. Using smaller images

D. Using more text

REFERENCE LABEL: NOT PART OF THE QUESTION ID S76-F2-Q1
Artificial Intelligence, Set 76: Presentations: structure, visuals and export · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.IMAGE.05 (local)

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Speaker notes say "studies show 90% of waste is recyclable." What should you do?

A. Delete all notes

B. Make the number bigger

C. Check the claim before saying it

D. Say it confidently, since speaker notes are written from verified sources by the tool

REFERENCE LABEL: NOT PART OF THE QUESTION ID S76-F2-Q2
Artificial Intelligence, Set 76: Presentations: structure, visuals and export · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.IMAGE.05 (local)

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An exported file replaced your font and lost an image. What kind of problem is this?

A. A spelling problem
B. A grounding problem
C. A problem with the topic
D. An export problem

REFERENCE LABEL: NOT PART OF THE QUESTION ID S76-F2-Q3
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Why reorder and rewrite a generated outline?

A. So it fits your audience and time

B. So it becomes longer

C. Because teachers require rewriting

D. Because generated outlines always contain nothing useful and must be thrown away

REFERENCE LABEL: NOT PART OF THE QUESTION ID S76-F2-Q4
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About how many slides suit a 20-minute talk?

A. Exactly 1
B. About 10 to 20
C. About 200
D. About 80

REFERENCE LABEL: NOT PART OF THE QUESTION ID S76-F2-Q5
Artificial Intelligence, Set 76: Presentations: structure, visuals and export · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.IMAGE.05 (local)

Set 77: Accessible visuals

Outcome AI.IMAGE.06: Make visuals accessible with alt text, readable contrast and clear labels

Module: Creative: images, design and presentations

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
ACCESSIBLE visuals work for more people, including people who are blind or have low vision, color blindness or reading difficulties. ALT TEXT is a short written description attached to an image so screen readers can read it aloud: it says what matters in the image for its purpose ("Bar chart: sales doubled from March to June"), not "image.jpg." CONTRAST means text stands out clearly from its background, such as dark text on a light background. CLEAR LABELS put words on chart lines and parts instead of relying on color alone. AI can draft alt text, but you check it.

What is alt text?

A. A short description so screen readers can read an image aloud
B. A caption in bold
C. A hidden password
D. The file name of the image, such as "photo123.jpg,"

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Which alt text is best for a sales chart?

A. "chart.png"
B. "Bar chart: sales doubled from March to June."
C. "Image"
D. "A chart image that someone made for this report."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S77-M-Q2
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What does good contrast mean?

A. Text is very small
B. Text is in all capitals
C. Text uses many different bright colors
D. Text stands out clearly from its background

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Why put labels on chart lines instead of relying on color?

A. To hide the data
B. To make the chart smaller
C. Some viewers cannot tell the colors apart
D. Because colors are not allowed in any chart

REFERENCE LABEL: NOT PART OF THE QUESTION ID S77-M-Q4
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Can AI draft alt text?

A. Yes, but you check it

B. No, it cannot describe images

C. Yes, and its drafts never need checking, since tools describe images perfectly every time

D. Only for photos of food

REFERENCE LABEL: NOT PART OF THE QUESTION ID S77-M-Q5
Artificial Intelligence, Set 77: Accessible visuals · MAIN · Q5/5 · Outcome AI.IMAGE.06 (local)

Learn
Imagine calling a friend and describing a photo they cannot see. You would not say "It's a file called IMG_204." You would say what matters: "It's our team holding the trophy after winning the final." You skip unimportant details like the exact color of the wall. Alt text works the same way: a short spoken-style description of what matters in the image, for someone who cannot see it.

Saying "It's a file called IMG_204" is like what?

A. A chart label
B. Useless alt text
C. Good contrast
D. Excellent alt text

REFERENCE LABEL: NOT PART OF THE QUESTION ID S77-B1-Q1
Artificial Intelligence, Set 77: Accessible visuals · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.IMAGE.06 (local)

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"Our team holding the trophy after the final" is like what?

A. Poor contrast
B. A caption in bold
C. Good alt text
D. A file name

REFERENCE LABEL: NOT PART OF THE QUESTION ID S77-B1-Q2
Artificial Intelligence, Set 77: Accessible visuals · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.IMAGE.06 (local)

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Why skip the color of the wall?

A. Because walls are invisible
B. To make it longer
C. Because colors must never be mentioned in any description of any image at all
D. It is not what matters

REFERENCE LABEL: NOT PART OF THE QUESTION ID S77-B1-Q3
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Who is alt text for?

A. Search engines only
B. People who cannot see the image
C. Printers
D. Only the person who created the image

REFERENCE LABEL: NOT PART OF THE QUESTION ID S77-B1-Q4
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A good phone description is...

A. A file name
B. Only one word
C. Very long, listing every single thing that appears anywhere in the picture
D. Short and focused on what matters

REFERENCE LABEL: NOT PART OF THE QUESTION ID S77-B1-Q5
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Kim's slide uses yellow text on a white background. What should she change?

A. Use dark text on the light background
B. Remove the text
C. Add more yellow
D. Make the yellow text slightly larger

REFERENCE LABEL: NOT PART OF THE QUESTION ID S77-F1-Q1
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A line chart uses red and green lines with no labels. What is the best fix?

A. Use two slightly different shades of green for the lines so the chart looks more modern

B. Remove the legend

C. Label each line with words

D. Make the chart smaller

REFERENCE LABEL: NOT PART OF THE QUESTION ID S77-F1-Q2
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A tool drafts alt text "A group of people" for a photo of the science fair winners. What should you do?

A. Revise it to say who and why it matters

B. Add the file name

C. Keep it, since every description written by a tool is already as detailed as needed

D. Delete the alt text

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A newsletter image announces the event date only inside the picture. What helps screen reader users?

A. Move the picture lower
B. Add more pictures
C. Also give the date as real text
D. Make the date in the picture bigger

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Which alt text suits a decorative divider line?

A. "Important image"
B. Little or none, since it carries no information
C. The file name
D. A detailed paragraph describing the line's exact color

REFERENCE LABEL: NOT PART OF THE QUESTION ID S77-F1-Q5
Artificial Intelligence, Set 77: Accessible visuals · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.IMAGE.06 (local)

Learn
Generated alt text can be wrong or miss the point: it might describe a chart as "colorful shapes" without the data, or guess at things that are not there. Alt text should match the image's purpose; a decorative border may need none, while an important chart needs its key finding. Color-only signals, such as "red means late," fail for many viewers, so add words or symbols. Some screen readers can recognize text inside images, but not reliably, so give important text as real text or alt text.

Generated alt text for a chart says "colorful shapes." What is missing?

A. The file size

B. The font name

C. More colors, since describing the colors fully is the most important part of chart alt text

D. The data or key finding

REFERENCE LABEL: NOT PART OF THE QUESTION ID S77-F2-Q1
Artificial Intelligence, Set 77: Accessible visuals · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.IMAGE.06 (local)

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A schedule marks late buses in red only. What improves it?

A. Add the word "late" or a symbol

B. Remove the late buses

C. Use pink instead

D. Make the red brighter, which ensures every reader can see that a bus is running late

REFERENCE LABEL: NOT PART OF THE QUESTION ID S77-F2-Q2
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Generated alt text mentions "a dog in the corner," but there is no dog. What happened?

A. Alt text always includes animals
B. The image changed
C. The tool guessed something not there
D. The dog must be hidden somewhere

REFERENCE LABEL: NOT PART OF THE QUESTION ID S77-F2-Q3
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Should a publisher rely on screen-reader text recognition to make important text inside an image accessible?

A. No; provide real text or alt text
B. Yes, if the image is large
C. Yes; screen readers read all text in every picture
D. Yes, if the words are in color

REFERENCE LABEL: NOT PART OF THE QUESTION ID S77-F2-Q4
Artificial Intelligence, Set 77: Accessible visuals · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.IMAGE.06 (local)

Learn
Generated alt text can be wrong or miss the point: it might describe a chart as "colorful shapes" without the data, or guess at things that are not there. Alt text should match the image's purpose; a decorative border may need none, while an important chart needs its key finding. Color-only signals, such as "red means late," fail for many viewers, so add words or symbols. Some screen readers can recognize text inside images, but not reliably, so give important text as real text or alt text.

What should alt text match?

A. The image size
B. The image's purpose
C. The file name
D. The length of the page

REFERENCE LABEL: NOT PART OF THE QUESTION ID S77-F2-Q5
Artificial Intelligence, Set 77: Accessible visuals · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.IMAGE.06 (local)

Set 78: Transcription versus speech generation

Outcome AI.MEDIA.01: Tell speech-to-text (transcription) from text-to-speech (generation) and check transcripts for errors and speaker labels

Module: Creative: audio, music, video and synthetic media

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
Two audio tasks run in opposite directions. TRANSCRIPTION, also called SPEECH-TO-TEXT, turns spoken audio into written words, for example meeting notes or captions. SPEECH GENERATION, also called TEXT-TO-SPEECH, turns written words into spoken audio, for example a voice reading an article aloud. Transcripts can contain errors: misheard words ("wreck a nice beach" for "recognize speech"), wrong names, missing words when people talk over each other, and wrong SPEAKER LABELS that say who said what. Before relying on a transcript, check names, numbers and key statements against the recording.

What does transcription do?

A. Deletes background noise only
B. Turns spoken audio into written words
C. Translates text into music
D. Turns written words into a spoken voice

REFERENCE LABEL: NOT PART OF THE QUESTION ID S78-M-Q1
Artificial Intelligence, Set 78: Transcription versus speech generation · MAIN · Q1/5 · Outcome AI.MEDIA.01 (local)

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What does text-to-speech do?

A. Makes video from text
B. Checks spelling
C. Writes down what people say in a meeting
D. Turns written words into spoken audio

REFERENCE LABEL: NOT PART OF THE QUESTION ID S78-M-Q2
Artificial Intelligence, Set 78: Transcription versus speech generation · MAIN · Q2/5 · Outcome AI.MEDIA.01 (local)

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What is a speaker label?

A. A note saying who said each part
B. The file name
C. The volume level
D. A sticker on a loudspeaker

REFERENCE LABEL: NOT PART OF THE QUESTION ID S78-M-Q3
Artificial Intelligence, Set 78: Transcription versus speech generation · MAIN · Q3/5 · Outcome AI.MEDIA.01 (local)

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Prepared example (a fictional meeting transcript)

Speaker 1: Our budget for the trip is fifteen hundred dollars.
Speaker 2: We can't book the bus until Friday.
Speaker 1: Then let's ask Mr. Patel to call the company.

In this prepared example, which detail is most important to check against the recording?

A. The punctuation only

B. The word "Then," because short words at the start of sentences are the most often wrong

C. "Fifteen hundred dollars" and "can't"

D. The word "the"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S78-M-Q4
Artificial Intelligence, Set 78: Transcription versus speech generation · MAIN · Q4/5 · Outcome AI.MEDIA.01 (local)

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"Wreck a nice beach" for "recognize speech" is what kind of error?

A. A misheard phrase
B. A missing word
C. A text-to-speech error
D. A speaker label error

REFERENCE LABEL: NOT PART OF THE QUESTION ID S78-M-Q5
Artificial Intelligence, Set 78: Transcription versus speech generation · MAIN · Q5/5 · Outcome AI.MEDIA.01 (local)

Learn
A COURT REPORTER listens to people talking and types every word: sound in, text out. That is like transcription. An audiobook NARRATOR reads a written book aloud: text in, sound out. That is like speech generation. A court reporter can mishear a name or mix up who spoke, which is why records are checked. A narrator can mispronounce a word, but the book's text stays the same.

A court reporter is like which task?

A. Translation
B. Image generation
C. Transcription
D. Speech generation

REFERENCE LABEL: NOT PART OF THE QUESTION ID S78-B1-Q1
Artificial Intelligence, Set 78: Transcription versus speech generation · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.MEDIA.01 (local)

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An audiobook narrator is like which task?

A. Editing
B. Summarizing
C. Transcription
D. Speech generation

REFERENCE LABEL: NOT PART OF THE QUESTION ID S78-B1-Q2
Artificial Intelligence, Set 78: Transcription versus speech generation · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.MEDIA.01 (local)

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What can a court reporter get wrong?

A. Nothing
B. A name or who spoke
C. The judge's salary
D. The color of the courtroom walls, which must be included in every court record

REFERENCE LABEL: NOT PART OF THE QUESTION ID S78-B1-Q3
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For transcription, what goes in?

A. Numbers only
B. Pictures
C. Sound
D. Written text

REFERENCE LABEL: NOT PART OF THE QUESTION ID S78-B1-Q4
Artificial Intelligence, Set 78: Transcription versus speech generation · BACKWARD 1 (concept simplification) · Q4/5 · Outcome AI.MEDIA.01 (local)

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For speech generation, what comes out?

A. Written text that matches the words someone said in a recorded meeting earlier

B. A spreadsheet

C. A picture

D. Sound

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Jo wants captions for a class video. Which task does she need?

A. Image generation
B. Transcription
C. Music generation
D. Text-to-speech

REFERENCE LABEL: NOT PART OF THE QUESTION ID S78-F1-Q1
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A blind student wants a news article read aloud. Which task fits?

A. Text-to-speech
B. Image tagging
C. Video editing
D. Transcription

REFERENCE LABEL: NOT PART OF THE QUESTION ID S78-F1-Q2
Artificial Intelligence, Set 78: Transcription versus speech generation · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.MEDIA.01 (local)

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A transcript says the meeting agreed to "cancel" the event, but you recall "continue." What should you do?

A. Ask attendees to vote again

B. Trust the transcript, since automatic transcripts never confuse words that sound alike

C. Listen to that part of the recording

D. Delete the recording

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A transcript credits the coach with a parent's question. What kind of error is this?

A. A wrong speaker label
B. A missing word
C. A speech generation error
D. A misheard word

REFERENCE LABEL: NOT PART OF THE QUESTION ID S78-F1-Q4
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Which transcript items most need checking?

A. Greetings
B. Names, numbers and decisions
C. Background music
D. Filler words like "um" and "uh,"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S78-F1-Q5
Artificial Intelligence, Set 78: Transcription versus speech generation · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.MEDIA.01 (local)

Learn
Transcription errors grow with noise, accents the system heard less often in training, fast speech, technical words and crosstalk. A transcript that looks clean may have quietly dropped a sentence or swapped "can" with "can't." For anything that matters, such as meeting decisions, quotes for an article or medical instructions, listen to the important parts again. Also ask permission before recording people; rules about recording consent differ by place.

Which recording is most likely to produce transcript errors?

A. A short clear voicemail
B. A slow, scripted reading
C. A quiet room with one person speaking slowly
D. A noisy room with people talking over each other

REFERENCE LABEL: NOT PART OF THE QUESTION ID S78-F2-Q1
Artificial Intelligence, Set 78: Transcription versus speech generation · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.MEDIA.01 (local)

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Why can "can" and "can't" be risky in transcripts?

A. Because transcripts are not allowed to contain any short words with apostrophes

B. They sound alike but mean the opposite

C. Because they are rare words

D. Because they are spelled the same

REFERENCE LABEL: NOT PART OF THE QUESTION ID S78-F2-Q2
Artificial Intelligence, Set 78: Transcription versus speech generation · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.MEDIA.01 (local)

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A transcript of a medical appointment lists a medicine dose. What should you do?

A. Double the dose to be safe

B. Follow the dose as written, since transcripts of appointments are checked by doctors

C. Ignore the dose

D. Confirm it with the provider or the recording

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Before recording a group conversation, what should you do?

A. Ask the people for permission
B. Record only the loud people
C. Use a hidden phone
D. Record secretly

REFERENCE LABEL: NOT PART OF THE QUESTION ID S78-F2-Q4
Artificial Intelligence, Set 78: Transcription versus speech generation · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.MEDIA.01 (local)

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A transcript misspells technical terms in a science lecture. Why?

A. Because the file was too small

B. Because the lecturer must have spelled the words incorrectly while speaking them aloud

C. Technical words are harder to recognize

D. Because science is not allowed

REFERENCE LABEL: NOT PART OF THE QUESTION ID S78-F2-Q5
Artificial Intelligence, Set 78: Transcription versus speech generation · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.MEDIA.01 (local)

Set 79: Subtitles and translated audio

Outcome AI.MEDIA.02: Check generated subtitles and translated audio for timing, meaning and names

Module: Creative: audio, music, video and synthetic media

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

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SUBTITLES show spoken words as text on screen; generated subtitles come from transcription and sometimes translation. TRANSLATED AUDIO, or AI DUBBING, replaces speech with a voice in another language. Check three things. TIMING: does each line appear when it is spoken and stay long enough to read? MEANING: is the translation faithful, or did an idiom such as "break a leg" become nonsense? NAMES AND TERMS: are people, places and technical words right? A fluent-sounding dub can still change meaning, so a speaker of both languages should review anything important.

What are subtitles?

A. Spoken words shown as text on screen

B. Music under a video

C. A voice in a second language that replaces the speaker's voice in the original video

D. The video's title only

REFERENCE LABEL: NOT PART OF THE QUESTION ID S79-M-Q1
Artificial Intelligence, Set 79: Subtitles and translated audio · MAIN · Q1/5 · Outcome AI.MEDIA.02 (local)

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Which is a timing problem?

A. A wrong place name
B. A misspelled name
C. A line appears after the person has finished speaking
D. A line that uses a word in the wrong language

REFERENCE LABEL: NOT PART OF THE QUESTION ID S79-M-Q2
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Prepared example (fictional generated subtitles)

Time	Original (English)	Generated subtitle (Spanish)
0:05	Break a leg tonight!	¡Rómpete una pierna esta noche!
0:09	The show starts at 7.	El espectáculo empieza a las 7.

In this prepared example, which subtitle has a meaning problem?

A. Neither line
B. The 0:05 line
C. Both lines
D. The 0:09 line

REFERENCE LABEL: NOT PART OF THE QUESTION ID S79-M-Q3
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Why can a fluent-sounding dub still be a problem?

A. Because dubs remove the video
B. Because dubs are always too loud
C. Because fluent speech is always a sign
D. It can change the meaning

REFERENCE LABEL: NOT PART OF THE QUESTION ID S79-M-Q4
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Who should review important translated media?

A. Nobody, since translation tools are now as accurate as professional human translators in every case

B. The video's editor only

C. A speaker of both languages

D. Someone who speaks only one language

REFERENCE LABEL: NOT PART OF THE QUESTION ID S79-M-Q5
Artificial Intelligence, Set 79: Subtitles and translated audio · MAIN · Q5/5 · Outcome AI.MEDIA.02 (local)

Learn
For each subtitle line ask three questions. TIMING: does it appear when the words are spoken and stay at least long enough to read (often about one second per few words)? MEANING: does it say what the speaker meant, not just a word-for-word copy? NAMES: are names and terms spelled correctly? Example: a speaker says "Dr. Nguyen, it's raining cats and dogs," and the subtitle reads "Dr. Wen, many animals are falling," two seconds late. All three checks fail.

The subtitle appears two seconds late. Which check fails?

A. Timing
B. None
C. Names
D. Meaning

REFERENCE LABEL: NOT PART OF THE QUESTION ID S79-B1-Q1
Artificial Intelligence, Set 79: Subtitles and translated audio · BACKWARD 1 (underlying skill) · Q1/5 · Outcome AI.MEDIA.02 (local)

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"Many animals are falling" fails which check?

A. None
B. Names
C. Timing
D. Meaning

REFERENCE LABEL: NOT PART OF THE QUESTION ID S79-B1-Q2
Artificial Intelligence, Set 79: Subtitles and translated audio · BACKWARD 1 (underlying skill) · Q2/5 · Outcome AI.MEDIA.02 (local)

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"Dr. Wen" for "Dr. Nguyen" fails which check?

A. None
B. Names
C. Timing
D. Meaning

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How many checks fail in the example?

A. Two
B. None
C. One
D. Three

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What should "raining cats and dogs" become?

A. A phrase meaning "raining very hard"

B. Nothing at all

C. "It is sunny"

D. A word-for-word version that keeps the cats and the dogs, so the subtitle is exact

REFERENCE LABEL: NOT PART OF THE QUESTION ID S79-B1-Q5
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A school posts a video with generated Spanish subtitles for families. What is the best step before posting?

A. Shorten the video

B. Have a Spanish speaker review them

C. Post right away, since every family will understand the subtitles even with small errors

D. Remove the English audio

REFERENCE LABEL: NOT PART OF THE QUESTION ID S79-F1-Q1
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Subtitles flash by too fast to read. Which check failed?

A. None
B. Names
C. Timing
D. Meaning

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Artificial Intelligence, Set 79: Subtitles and translated audio · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.MEDIA.02 (local)

Learn
SUBTITLES show spoken words as text on screen; generated subtitles come from transcription and sometimes translation. TRANSLATED AUDIO, or AI DUBBING, replaces speech with a voice in another language. Check three things. TIMING: does each line appear when it is spoken and stay long enough to read? MEANING: is the translation faithful, or did an idiom such as "break a leg" become nonsense? NAMES AND TERMS: are people, places and technical words right? A fluent-sounding dub can still change meaning, so a speaker of both languages should review anything important.

A dub translates "Mrs. Kowalski" as "Mrs. Cow." What failed?

A. Timing, since the dubbed name was said slightly later than in the original video
B. Only the volume
C. Nothing
D. Names

REFERENCE LABEL: NOT PART OF THE QUESTION ID S79-F1-Q3
Artificial Intelligence, Set 79: Subtitles and translated audio · FORWARD 1: APPLY IT · Q3/5 · Outcome AI.MEDIA.02 (local)

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A subtitle says "5 miles" but the translated audio says "5 kilometers." What should you do?

A. Delete the numbers

B. Check which is right and fix the other

C. Use the louder one

D. Leave both, since miles and kilometers are about the same distance for most purposes

REFERENCE LABEL: NOT PART OF THE QUESTION ID S79-F1-Q4
Artificial Intelligence, Set 79: Subtitles and translated audio · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.MEDIA.02 (local)

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Which check is most important for a safety video?

A. The video length
B. The music volume
C. Meaning of each instruction
D. The font of the subtitles

REFERENCE LABEL: NOT PART OF THE QUESTION ID S79-F1-Q5
Artificial Intelligence, Set 79: Subtitles and translated audio · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.MEDIA.02 (local)

Learn
Translation can shift tone and politeness as well as words: a friendly phrase may become rude, or a formal one too casual. Jokes, numbers and units (miles versus kilometers) need care. Dubbed voices that imitate the original speaker raise consent questions. When you publish translated media, say that subtitles or audio were machine-generated if a human did not fully review them, and give viewers a way to report errors.

A polite request in the original became a rude command in the dub. What changed?

A. The tone
B. The speaker's name
C. Nothing important
D. The timing

REFERENCE LABEL: NOT PART OF THE QUESTION ID S79-F2-Q1
Artificial Intelligence, Set 79: Subtitles and translated audio · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.MEDIA.02 (local)

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A dub uses a voice that imitates the original speaker. What question does this raise?

A. Whether the video has color
B. Whether subtitles are needed
C. Whether the dub is long enough
D. Whether the speaker consented

REFERENCE LABEL: NOT PART OF THE QUESTION ID S79-F2-Q2
Artificial Intelligence, Set 79: Subtitles and translated audio · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.MEDIA.02 (local)

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A team publishes machine subtitles that no one fully reviewed. What should they add?

A. Nothing, since viewers always assume that every subtitle they read has been reviewed

B. More subtitles

C. A note that subtitles were machine-generated

D. A longer title

REFERENCE LABEL: NOT PART OF THE QUESTION ID S79-F2-Q3
Artificial Intelligence, Set 79: Subtitles and translated audio · FORWARD 2: HANDLE THE COMPLICATIONS · Q3/5 · Outcome AI.MEDIA.02 (local)

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Why give viewers a way to report errors?

A. So mistakes can be found and fixed
B. Because it is required for every video
C. To collect email addresses
D. So viewers can rewrite the entire video

REFERENCE LABEL: NOT PART OF THE QUESTION ID S79-F2-Q4
Artificial Intelligence, Set 79: Subtitles and translated audio · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.MEDIA.02 (local)

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A joke in the original falls flat in translation. What is the best approach?

A. Remove the whole scene

B. Have a reviewer adapt it

C. Add laughter sounds

D. Translate it word for word anyway, since the exact words matter more than the joke

REFERENCE LABEL: NOT PART OF THE QUESTION ID S79-F2-Q5
Artificial Intelligence, Set 79: Subtitles and translated audio · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.MEDIA.02 (local)

Set 80: Voice cloning and consent

Outcome AI.MEDIA.03: Explain voice cloning and why consent and disclosure matter, using dated legal examples

Module: Creative: audio, music, video and synthetic media

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
VOICE CLONING uses recordings of a real person to make a synthetic voice that sounds like them saying new words. It has real uses, such as helping people who are losing their voice, but it can also fool listeners: scam calls, fake endorsements, or false statements. CONSENT (the person agrees) and DISCLOSURE (listeners are told the voice is synthetic) matter. Some laws now address this. Tennessee's ELVIS Act (effective July 1, 2024) added a person's voice, including a simulated one, to that state's publicity rights. In February 2024, the US Federal Communications Commission (FCC) ruled that AI-generated voices count as "artificial" voices, so covered calls follow the same consent rules, and exceptions, as other robocalls.

What is voice cloning?

A. Recording someone's voice once and playing back exactly the same recording many times

B. Making music from noise

C. Making a synthetic voice that sounds like a real person

D. Turning speech into text

REFERENCE LABEL: NOT PART OF THE QUESTION ID S80-M-Q1
Artificial Intelligence, Set 80: Voice cloning and consent · MAIN · Q1/5 · Outcome AI.MEDIA.03 (local)

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Which is a helpful use of voice cloning?

A. Helping someone who is losing their voice
B. Faking a politician's statement
C. Making a fake celebrity endorsement
D. Calling a grandparent

REFERENCE LABEL: NOT PART OF THE QUESTION ID S80-M-Q2
Artificial Intelligence, Set 80: Voice cloning and consent · MAIN · Q2/5 · Outcome AI.MEDIA.03 (local)

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What does disclosure mean here?

A. Keeping secret that the voice was made by a computer so the audio sounds more natural

B. Telling listeners the voice is synthetic

C. Making the audio louder

D. Publishing the person's address

REFERENCE LABEL: NOT PART OF THE QUESTION ID S80-M-Q3
Artificial Intelligence, Set 80: Voice cloning and consent · MAIN · Q3/5 · Outcome AI.MEDIA.03 (local)

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What did Tennessee's ELVIS Act (effective July 1, 2024) do?

A. Required all songs to be written by humans
B. Made voice cloning legal everywhere
C. Banned every use of synthetic voices anywhere
D. Added voice to that state's publicity rights

REFERENCE LABEL: NOT PART OF THE QUESTION ID S80-M-Q4
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What did the FCC rule in February 2024?

A. AI voices may be used in any phone call to anyone
B. Phone calls must be recorded
C. AI voices in calls count as "artificial" voices
D. Voice cloning is a crime everywhere

REFERENCE LABEL: NOT PART OF THE QUESTION ID S80-M-Q5
Artificial Intelligence, Set 80: Voice cloning and consent · MAIN · Q5/5 · Outcome AI.MEDIA.03 (local)

Learn
Your SIGNATURE stands for you: when people see it, they believe you agreed. Copying someone's signature without permission to sign a document would be wrong, even if the copy is perfect. A cloned VOICE works similarly: listeners believe the person is speaking. Using it with their CONSENT, and saying it is synthetic, is like signing with permission and a note. Using it without consent to fool people is like forging a signature.

Why does a signature matter?

A. Because it is written in ink

B. Because signatures are always the most beautiful part of any written document

C. Because it shows your height

D. People believe you agreed when they see it

REFERENCE LABEL: NOT PART OF THE QUESTION ID S80-B1-Q1
Artificial Intelligence, Set 80: Voice cloning and consent · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.MEDIA.03 (local)

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Copying a signature without permission is like what?

A. Writing your own name
B. Typing a letter
C. Cloning a voice without consent
D. Getting written permission

REFERENCE LABEL: NOT PART OF THE QUESTION ID S80-B1-Q2
Artificial Intelligence, Set 80: Voice cloning and consent · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.MEDIA.03 (local)

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Using a cloned voice with consent and disclosure is like what?

A. Signing with permission and a note
B. Erasing a signature
C. Losing a pen
D. Forging a signature perfectly

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Why do listeners trust a familiar voice?

A. Because it is recorded
B. They believe the person is speaking
C. Because it is loud
D. Because voices always tell the truth

REFERENCE LABEL: NOT PART OF THE QUESTION ID S80-B1-Q4
Artificial Intelligence, Set 80: Voice cloning and consent · BACKWARD 1 (concept simplification) · Q4/5 · Outcome AI.MEDIA.03 (local)

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Is a perfect copy acceptable without permission?

A. No; permission is what matters

B. Yes, if it is short

C. Yes, because a copy that is perfect cannot cause any harm to the person whose voice it is

D. Only on weekends

REFERENCE LABEL: NOT PART OF THE QUESTION ID S80-B1-Q5
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A podcaster wants to use a cloned voice of a guest to fix a few words. What should come first?

A. A louder microphone

B. A longer episode

C. The guest's clear permission

D. Nothing, since small fixes to a few words never need anyone's permission at all

REFERENCE LABEL: NOT PART OF THE QUESTION ID S80-F1-Q1
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An ad uses a synthetic voice that sounds like a famous singer without permission. What is the concern?

A. The ad is too short
B. The ad has music
C. None, since a synthetic voice is a new voice
D. It uses the singer's identity without consent

REFERENCE LABEL: NOT PART OF THE QUESTION ID S80-F1-Q2
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A school announcement uses a synthetic voice reading the principal's message, with permission. What should it add?

A. Nothing, since permission from the principal means listeners never need to be told anything

B. A different message

C. A note that the voice is synthetic

D. Louder music

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Grandpa gets a call in his grandson's voice asking for money urgently. What should he do?

A. Give his bank details
B. Hang up and call his grandson on a known number
C. Send the money right away, since the voice sounded exactly like his grandson
D. Call back the number that called

REFERENCE LABEL: NOT PART OF THE QUESTION ID S80-F1-Q4
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A company makes non-emergency telemarketing robocalls to cellphones with an AI voice and no prior consent. Under the FCC's 2024 ruling, what applies?

A. A ban on all phone calls

B. Only a fee

C. No rules, because AI voices were left out of phone-call rules entirely by the ruling

D. The same consent rules as other artificial-voice calls

REFERENCE LABEL: NOT PART OF THE QUESTION ID S80-F1-Q5
Artificial Intelligence, Set 80: Voice cloning and consent · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.MEDIA.03 (local)

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Laws on voices differ by place and are still developing, so a use allowed in one state or country may not be allowed in another, and courts are still deciding many questions. Practical rules hold everywhere: get clear written permission before cloning someone's voice, say when audio is synthetic, never imitate a real person to deceive, and be suspicious of urgent calls that "sound like" a relative asking for money. Families can agree on a code word to verify each other on the phone.

A voice use is allowed in one state. Is it allowed everywhere?

A. Not necessarily; laws differ by place

B. Yes, if it is online

C. No, it is banned everywhere

D. Yes, because every state and every country follows exactly the same rules about voices

REFERENCE LABEL: NOT PART OF THE QUESTION ID S80-F2-Q1
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What is a practical rule everywhere?

A. Only clone famous people
B. Get clear permission before cloning a voice
C. Never tell anyone a voice is synthetic
D. Clone any voice you like

REFERENCE LABEL: NOT PART OF THE QUESTION ID S80-F2-Q2
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How can a family protect against fake voice calls?

A. Post their voices online

B. Answer every call quickly and do whatever the familiar voice asks without any delay

C. Agree on a code word to verify each other

D. Stop using phones

REFERENCE LABEL: NOT PART OF THE QUESTION ID S80-F2-Q3
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Are all legal questions about voice cloning settled?

A. No; courts are still deciding many

B. There are no laws about voices

C. Yes, by the ELVIS Act for the whole world

D. Yes; every possible question about synthetic voices was answered by courts years ago

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Which use most clearly crosses the line?

A. A clearly labeled synthetic narrator
B. A screen reader voice
C. Using a synthetic voice, with consent
D. Imitating a real person to deceive listeners

REFERENCE LABEL: NOT PART OF THE QUESTION ID S80-F2-Q5
Artificial Intelligence, Set 80: Voice cloning and consent · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.MEDIA.03 (local)

Set 81: Music generation and rights

Outcome AI.MEDIA.04: Check the rights and terms that apply before using generated music

Module: Creative: audio, music, video and synthetic media

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
MUSIC GENERATORS create songs, beats or background tracks from text descriptions. Before using generated music, check the RIGHTS. (1) The tool's TERMS: may you use it commercially, for example in a monetized video? Is credit required? Some free plans allow only personal use. (2) COPYRIGHT: in the United States, the Copyright Office's January 2025 report said that prompts alone do not make you the author of the output; your own lyrics, melodies or meaningful edits can be protected. (3) SIMILARITY: if a track sounds very like an existing song or imitates a specific artist's voice, avoid it. Keep a record of the tool, plan and date.

What should you check first in a music tool's terms?

A. The tool's logo
B. The number of users
C. Whether commercial use is allowed
D. The color scheme of the app

REFERENCE LABEL: NOT PART OF THE QUESTION ID S81-M-Q1
Artificial Intelligence, Set 81: Music generation and rights · MAIN · Q1/5 · Outcome AI.MEDIA.04 (local)

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Per the US Copyright Office's January 2025 report, do prompts alone make you the author?

A. Yes, because writing a detailed prompt always makes you the full author of every output

B. No

C. Only for songs under 3 minutes

D. Only for long prompts

REFERENCE LABEL: NOT PART OF THE QUESTION ID S81-M-Q2
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Which can be protected by copyright, per that report?

A. Your own lyrics or meaningful edits
B. Any random sound
C. Someone else's melody
D. The tool's output, in full, simply

REFERENCE LABEL: NOT PART OF THE QUESTION ID S81-M-Q3
Artificial Intelligence, Set 81: Music generation and rights · MAIN · Q3/5 · Outcome AI.MEDIA.04 (local)

Learn
MUSIC GENERATORS create songs, beats or background tracks from text descriptions. Before using generated music, check the RIGHTS. (1) The tool's TERMS: may you use it commercially, for example in a monetized video? Is credit required? Some free plans allow only personal use. (2) COPYRIGHT: in the United States, the Copyright Office's January 2025 report said that prompts alone do not make you the author of the output; your own lyrics, melodies or meaningful edits can be protected. (3) SIMILARITY: if a track sounds very like an existing song or imitates a specific artist's voice, avoid it. Keep a record of the tool, plan and date.

A generated track sounds almost exactly like a famous song. What should you do?

A. Speed it up slightly
B. Change the title
C. Use it
D. Avoid using it

REFERENCE LABEL: NOT PART OF THE QUESTION ID S81-M-Q4
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What record should you keep?

A. Nothing
B. The tool, plan and date
C. The tool's home page design
D. Only the song's title

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Artificial Intelligence, Set 81: Music generation and rights · MAIN · Q5/5 · Outcome AI.MEDIA.04 (local)

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What rules come with a library book?

A. You can read it but not sell copies
B. You must memorize it
C. You must read it in one day
D. You can do anything at all with it

REFERENCE LABEL: NOT PART OF THE QUESTION ID S81-B1-Q1
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In music tools, what sets the rules?

A. The song's length
B. The weather
C. The listener's mood
D. The tool's terms

REFERENCE LABEL: NOT PART OF THE QUESTION ID S81-B1-Q2
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Writing your name in a library book is like what?

A. Composing the music yourself from scratch, note by note, without any tool at all

B. Buying the book

C. Typing a prompt

D. Returning the book

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A free plan allowing only personal use is like what?

A. Owning the library
B. Writing a new book
C. Photocopying the library book
D. Reading a library book at home

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What is the lesson?

A. Check the rules before using generated music
B. Generated music has no rules at all, so it can be used anywhere for anything
C. Never listen to music
D. Only use library music

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Eli wants generated background music for a monetized video. What should he check?

A. Nothing, because all generated music may be used in any video, including monetized ones

B. Only the song length

C. That his plan allows commercial use

D. Only the tempo

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A student writes her own lyrics and uses a tool for the beat. Which part is most clearly hers?

A. The tool's code
B. The lyrics
C. Neither
D. The generated beat

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A prompt asks for a song "in the voice of" a named living singer. What is the concern?

A. The song may be too slow
B. The song may be too short
C. None
D. Imitating a real artist

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A tool's terms say "credit required." What should you do?

A. Credit yourself only
B. Change tools after publishing
C. Include the credit as described
D. Leave out the credit

REFERENCE LABEL: NOT PART OF THE QUESTION ID S81-F1-Q4
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Why save proof of your plan when you generated a track?

A. Because it is required by every country
B. Terms can change later
C. Because tools delete every song that was made on a free plan after one week
D. To make the song louder

REFERENCE LABEL: NOT PART OF THE QUESTION ID S81-F1-Q5
Artificial Intelligence, Set 81: Music generation and rights · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.MEDIA.04 (local)

Learn
Rights questions are still being argued in courts and agencies, and answers differ by country, so this course does not settle them. Practical steps reduce risk: read the current terms, not old ones; keep proof of your plan when you generated the track; avoid prompts like "in the voice of" a named singer; and for school or public events, follow the venue's rules on music. Platforms may also remove or flag content whose music matches existing recordings.

Are rights questions about generated music fully settled?

A. No; they are still being argued
B. There are no rules
C. Yes, the same in every country
D. Yes; one court case

REFERENCE LABEL: NOT PART OF THE QUESTION ID S81-F2-Q1
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Which terms should you rely on?

A. Any terms you like

B. A friend's memory of the terms

C. The current terms

D. Terms from a blog post written several years ago about a different version of the tool

REFERENCE LABEL: NOT PART OF THE QUESTION ID S81-F2-Q2
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A platform flags your video because its generated music matches an existing recording. What does this show?

A. That videos need no music

B. Generated music can resemble existing works

C. That the platform is always wrong, since generated songs cannot match existing recordings

D. That music is banned

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For a school concert, what should guide the music choice?

A. The loudest song
B. Whatever is trending
C. Only what the tool allows
D. The venue's rules on music

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What does this course do on disputed music rights?

A. It does not settle them
B. It says all generated music is illegal
C. It gives one final answer that applies in every country for every kind of song
D. It says anything goes

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Artificial Intelligence, Set 81: Music generation and rights · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.MEDIA.04 (local)

Set 82: Generated and edited video

Outcome AI.MEDIA.05: Check generated or edited video for consistency over time and disclose synthetic presenters

Module: Creative: audio, music, video and synthetic media

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
Generated and AI-edited VIDEO can show people, places and events that never happened, or change real footage. Video adds a challenge that single images lack: CONSISTENCY OVER TIME. Check frame to frame: do faces, clothes, objects and backgrounds stay the same? Do hands, text and logos warp? Does lip movement match the words? Does movement obey ordinary physics, such as objects not passing through each other? When a video uses a SYNTHETIC PRESENTER, an AI-generated person or avatar, tell viewers clearly, for example with an on-screen label or a note in the description.

What challenge does video add compared with single images?

A. Consistency over time

B. Videos have no people

C. Videos have no colors

D. Video can never be generated by AI tools, so it is always real footage from a camera

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Which is a sign of a generated video?

A. A video with music
B. A short video
C. A shirt that changes pattern between frames
D. A video that was recorded in a bright room

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What should you check about lip movement?

A. Whether the person has a microphone
B. Whether it matches the words
C. Whether the person is smiling
D. Whether the lips are a natural color

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What is a synthetic presenter?

A. A microphone
B. A video file format
C. A real news anchor who reads the news from a teleprompter
D. An AI-generated person or avatar presenting a video

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How should a synthetic presenter be disclosed?

A. With an on-screen label or description note

B. It does not need disclosing if it looks realistic enough that viewers will not notice

C. By adding music

D. By making the video shorter

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Movie crews have a job called CONTINUITY: making sure that if an actor's coffee cup is half full in one shot, it is not suddenly full in the next, and that a bandage stays on the same arm. Viewers notice these slips. Generated video has continuity problems too, often more of them: a necklace disappears, a sign changes words, a person's hair changes length between moments.

What is the continuity job in movies?

A. Selling tickets

B. Choosing actors

C. Keeping details the same between shots

D. Writing the music that plays during the final scenes of the movie for the audience

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A coffee cup half full, then full in the next shot, is what?

A. Good editing
B. A lighting choice
C. A clever special effect
D. A continuity slip

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A necklace disappearing in a generated video is like what?

A. A sound error
B. A continuity slip
C. A subtitle error
D. Perfect continuity

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Who notices continuity slips?

A. Viewers
B. Only the director
C. Only the camera
D. Nobody

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What should you look for in generated video?

A. Only the length
B. Details changing between moments
C. Only the music
D. Only the title at the start

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A training video uses an AI avatar to present safety rules. What should it include?

A. Nothing extra, since viewers only care about the safety rules and not about who presents them

B. A fake name and job title

C. A clear note that the presenter is AI-generated

D. A claim the presenter is real

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In a clip, a person's earrings vanish and return. What does this suggest?

A. A bad microphone
B. A slow internet connection
C. Nothing
D. Possible generation or editing

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A video's words do not match the speaker's lip movements. What should you consider?

A. Nothing

B. That the video is very old

C. The audio may have been changed or generated

D. That the speaker has a cold, which always makes lips move differently from the words

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A ball passes through a wall in a video meant to look real. What does this break?

A. Ordinary physics
B. Lip sync, since the ball and the wall are both moving their lips in the scene
C. Subtitles
D. Nothing

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A small business makes an ad with a synthetic spokesperson. What is required by honesty?

A. Using more music
B. Hiding the business name
C. Saying the spokesperson is a real customer
D. Telling viewers the spokesperson is synthetic

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Some edits are subtle: a real video with one changed word, a background replaced, or a crowd made larger. Even honest edits can mislead if context is removed, such as a clip cut so a person seems to say the opposite. When you publish edited or generated video, disclose the changes that matter, keep the original footage, and never make a real person appear to say or endorse something without approval, or present it as genuine. If a synthetic presenter looks like a real person, you need that person's consent.

A real video has one word changed by AI. Why can this matter?

A. It changes the colors
B. It can change the meaning
C. It makes the file bigger
D. It cannot matter

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A clip is cut so a person seems to say the opposite of what they meant. What is the problem?

A. Removed context misleads
B. The clip is too long
C. The clip has no music
D. No problem

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Artificial Intelligence, Set 82: Generated and edited video · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.MEDIA.05 (local)

Learn
Some edits are subtle: a real video with one changed word, a background replaced, or a crowd made larger. Even honest edits can mislead if context is removed, such as a clip cut so a person seems to say the opposite. When you publish edited or generated video, disclose the changes that matter, keep the original footage, and never make a real person appear to say or endorse something without approval, or present it as genuine. If a synthetic presenter looks like a real person, you need that person's consent.

When publishing an edited video, what should you keep?

A. Nothing
B. The original footage
C. Only the audio
D. Only the edited version

REFERENCE LABEL: NOT PART OF THE QUESTION ID S82-F2-Q3
Artificial Intelligence, Set 82: Generated and edited video · FORWARD 2: HANDLE THE COMPLICATIONS · Q3/5 · Outcome AI.MEDIA.05 (local)

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A synthetic presenter looks like a real local teacher. What is needed?

A. A different background
B. A longer video
C. Nothing
D. The teacher's consent

REFERENCE LABEL: NOT PART OF THE QUESTION ID S82-F2-Q4
Artificial Intelligence, Set 82: Generated and edited video · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.MEDIA.05 (local)

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Which use should never happen?

A. Replacing a plain background with a nicer one and noting the change in the video description

B. Adding captions

C. Trimming silence

D. Making a real person seem to endorse a product without approval

REFERENCE LABEL: NOT PART OF THE QUESTION ID S82-F2-Q5
Artificial Intelligence, Set 82: Generated and edited video · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.MEDIA.05 (local)

Set 83: Deepfakes: checking what you see and hear

Outcome AI.MEDIA.06: Verify surprising images, audio or video by checking the source and context, not one visual oddity

Module: Creative: audio, music, video and synthetic media

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
A DEEPFAKE is realistic generated or altered media that shows someone saying or doing something they did not. Spotting one by a single visual oddity, such as strange blinking or blurry ears, is unreliable: real videos have glitches too, and newer fakes may have none. Better checks look at SOURCE and CONTEXT. Who first posted it? Do trusted news outlets or the person's official accounts report it? Does a reverse image search find older, original versions? Does it fit known facts about time and place? A surprising clip that makes you angry or scared deserves the most checking.

What is a deepfake?

A. A cartoon
B. A very old photo
C. Any video that has been edited at all, including trimming silence
D. Realistic fake media showing someone doing what they did not

REFERENCE LABEL: NOT PART OF THE QUESTION ID S83-M-Q1
Artificial Intelligence, Set 83: Deepfakes: checking what you see and hear · MAIN · Q1/5 · Outcome AI.MEDIA.06 (local)

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Why is looking for one visual oddity unreliable?

A. Because oddities are always invisible

B. Real videos have glitches and new fakes may have none

C. Because all videos are fake

D. Because fakes always show strange blinking, so if blinking looks normal the video is real

REFERENCE LABEL: NOT PART OF THE QUESTION ID S83-M-Q2
Artificial Intelligence, Set 83: Deepfakes: checking what you see and hear · MAIN · Q2/5 · Outcome AI.MEDIA.06 (local)

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Which is a source check?

A. Looking at the thumbnail

B. Counting how many times the person in the video blinks during the first ten seconds

C. Finding who first posted it

D. Checking the video length

REFERENCE LABEL: NOT PART OF THE QUESTION ID S83-M-Q3
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What can a reverse image search find?

A. Older, original versions of an image

B. The video's future views

C. The camera's battery level

D. The exact person who made a fake, along with their home address and phone number

REFERENCE LABEL: NOT PART OF THE QUESTION ID S83-M-Q4
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Which clip deserves the most checking?

A. A weather report

B. A surprising one that makes you angry or scared

C. A calm, ordinary clip of a local park on a sunny afternoon with nothing much happening

D. A cooking video

REFERENCE LABEL: NOT PART OF THE QUESTION ID S83-M-Q5
Artificial Intelligence, Set 83: Deepfakes: checking what you see and hear · MAIN · Q5/5 · Outcome AI.MEDIA.06 (local)

Learn
Before believing or sharing a surprising clip, ask four questions. (1) SOURCE: who posted it first, and are they trustworthy? (2) CONFIRMATION: do trusted outlets or the person's official accounts report it? (3) EARLIER VERSIONS: does a reverse image search find an original that is different? (4) CONTEXT: do the date, place and weather fit? Example: a "live" video of snow in a desert city on a day that was 35 °C, posted by a brand-new account, fails questions 1 and 4.

A clip posted by a brand-new anonymous account fails which question?

A. Earlier versions
B. Confirmation
C. Source
D. Context

REFERENCE LABEL: NOT PART OF THE QUESTION ID S83-B1-Q1
Artificial Intelligence, Set 83: Deepfakes: checking what you see and hear · BACKWARD 1 (underlying skill) · Q1/5 · Outcome AI.MEDIA.06 (local)

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Snow on a 35 °C day fails which question?

A. Context
B. Confirmation
C. Earlier versions
D. Source

REFERENCE LABEL: NOT PART OF THE QUESTION ID S83-B1-Q2
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No trusted outlet reports a "major" event. Which question fails?

A. Context
B. Source
C. Earlier versions
D. Confirmation

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A reverse search finds the same photo from 2018 with a different caption. Which question?

A. Context
B. Earlier versions
C. Source
D. Confirmation

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How many of the four questions does the snow example fail?

A. One
B. None
C. Four
D. Two

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A video shows a mayor announcing schools will close tomorrow. What should you check first?

A. The video's music
B. How many likes it has
C. The city's official accounts or trusted news
D. Whether the mayor blinks normally in the video

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A dramatic photo of a flood is going viral. A reverse search finds it from another country years ago. What does this show?

A. It is being shared with false context
B. Nothing
C. That the photo is AI-generated
D. That the flood is happening again

REFERENCE LABEL: NOT PART OF THE QUESTION ID S83-F1-Q2
Artificial Intelligence, Set 83: Deepfakes: checking what you see and hear · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.MEDIA.06 (local)

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A friend shares a shocking audio clip of a celebrity. The clip has no source. What is the best response?

A. Report the friend to police
B. Do not share it until it is confirmed
C. Add your own comments and repost
D. Share it with a note saying "might be fake,"

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Which detail is the strongest sign a clip is genuine?

A. It was posted recently
B. It has many views
C. The clip looks very high quality and sharp
D. Several trusted outlets confirm it independently

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A clip shows a protest "today," but trees are bare and it is summer. Which check does this use?

A. Earlier versions
B. Confirmation
C. Context
D. Source

REFERENCE LABEL: NOT PART OF THE QUESTION ID S83-F1-Q5
Artificial Intelligence, Set 83: Deepfakes: checking what you see and hear · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.MEDIA.06 (local)

Learn
Two mistakes are common. One is believing a fake because it looks real. The other is dismissing real evidence as "probably AI," which lets people deny things they really did. Both are avoided by checking sources rather than trusting looks. Some deepfakes cause serious harm, such as nonconsensual intimate images. In the United States, the TAKE IT DOWN Act requires covered platforms, from May 19, 2026, to offer a removal process and act on valid requests within 48 hours. Report such content, and do not share it.

What are the two common mistakes?

A. Believing fakes and dismissing real evidence as AI

B. Using reverse search and reading captions

C. Watching videos and reading articles

D. Checking sources too carefully and asking trusted news outlets whether events really happened

REFERENCE LABEL: NOT PART OF THE QUESTION ID S83-F2-Q1
Artificial Intelligence, Set 83: Deepfakes: checking what you see and hear · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.MEDIA.06 (local)

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Someone says a real recording of them is "probably AI" to avoid blame. What is this an example of?

A. Disclosure
B. Dismissing real evidence
C. Good fact-checking
D. Believing a fake

REFERENCE LABEL: NOT PART OF THE QUESTION ID S83-F2-Q2
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What avoids both mistakes?

A. Checking sources rather than trusting looks

B. Trusting your first impression of how real a video looks, without any further checks

C. Never watching videos

D. Believing whatever is most popular

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From May 19, 2026, what must covered US platforms do under the TAKE IT DOWN Act?

A. Nothing new
B. Ban all video uploads
C. Offer a removal process and act on valid requests within 48 hours
D. Remove every AI-generated image from their sites, including art

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You receive a nonconsensual intimate deepfake of a classmate. What should you do?

A. Keep it to show later
B. Post it with a warning
C. Forward it to a few friends
D. Report it and do not share it

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Artificial Intelligence, Set 83: Deepfakes: checking what you see and hear · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.MEDIA.06 (local)

Set 84: Provenance labels and the limits of detection

Outcome AI.MEDIA.07: Explain content credentials and watermarks and why missing labels or detector scores are not proof

Module: Creative: audio, music, video and synthetic media

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Learn
PROVENANCE means a record of where content came from and how it was changed. CONTENT CREDENTIALS, based on the C2PA standard (Coalition for Content Provenance and Authenticity), attach signed information to a file, such as which camera or tool made it and what edits were applied. WATERMARKS are hidden or visible marks some generators add to their output. These help, but they have LIMITS. Records can be stripped by screenshots or re-uploading. A MISSING label does not prove content is real or fake. DETECTOR tools give scores, often shown as percentages, and make errors both ways.

What does provenance mean?

A. A record of where content came from and how it changed

B. The number of views

C. A rating of how beautiful or well made a picture is compared with other pictures online

D. The file size

REFERENCE LABEL: NOT PART OF THE QUESTION ID S84-M-Q1
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What are Content Credentials based on?

A. A social media app
B. A camera brand
C. The C2PA standard
D. A single government law

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How can provenance records be lost?

A. By looking at it

B. By screenshots or re-uploading

C. By opening the file

D. They can never be lost once they are attached, no matter what happens to the file

REFERENCE LABEL: NOT PART OF THE QUESTION ID S84-M-Q3
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An image has no provenance label. What does that prove?

A. That it is real

B. That it is old

C. That the image must be fake, because every real photo always carries a provenance label

D. Nothing about whether it is real or fake

REFERENCE LABEL: NOT PART OF THE QUESTION ID S84-M-Q4
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What do detector tools give?

A. Certain answers that prove beyond doubt
B. Scores that can be wrong both ways
C. A copyright license
D. The creator's name

REFERENCE LABEL: NOT PART OF THE QUESTION ID S84-M-Q5
Artificial Intelligence, Set 84: Provenance labels and the limits of detection · MAIN · Q5/5 · Outcome AI.MEDIA.07 (local)

Learn
A package TRACKING HISTORY shows where a parcel has been: shipped from the warehouse, scanned in your city, delivered. It helps you trust the package came from the right place. But if someone repacks the item in a plain box, the history is gone, and a box with no tracking is not necessarily stolen. Provenance records work similarly: helpful when present, easy to lose, and their absence proves little.

What does a tracking history show?

A. The driver's name only
B. The weather
C. What is inside the parcel
D. Where the parcel has been

REFERENCE LABEL: NOT PART OF THE QUESTION ID S84-B1-Q1
Artificial Intelligence, Set 84: Provenance labels and the limits of detection · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.MEDIA.07 (local)

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Provenance for a file is like what?

A. A tracking history
B. A gift card
C. A price tag
D. A brand-new plain box

REFERENCE LABEL: NOT PART OF THE QUESTION ID S84-B1-Q2
Artificial Intelligence, Set 84: Provenance labels and the limits of detection · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.MEDIA.07 (local)

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Repacking the item in a plain box is like what?

A. Delivering faster

B. Opening the parcel

C. Stripping provenance by re-uploading

D. Adding a new and more detailed tracking label that shows the parcel's full history

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A box with no tracking proves what?

A. That it is valuable
B. Not necessarily anything
C. That the box was definitely stolen and should be reported to the police at once
D. That it is empty

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Artificial Intelligence, Set 84: Provenance labels and the limits of detection · BACKWARD 1 (concept simplification) · Q4/5 · Outcome AI.MEDIA.07 (local)

Learn
A package TRACKING HISTORY shows where a parcel has been: shipped from the warehouse, scanned in your city, delivered. It helps you trust the package came from the right place. But if someone repacks the item in a plain box, the history is gone, and a box with no tracking is not necessarily stolen. Provenance records work similarly: helpful when present, easy to lose, and their absence proves little.

What is the lesson?

A. Only buy online
B. Tracking is useless
C. Provenance helps but its absence proves little
D. Never trust anything that has a tracking history

REFERENCE LABEL: NOT PART OF THE QUESTION ID S84-B1-Q5
Artificial Intelligence, Set 84: Provenance labels and the limits of detection · BACKWARD 1 (concept simplification) · Q5/5 · Outcome AI.MEDIA.07 (local)

Learn
PROVENANCE means a record of where content came from and how it was changed. CONTENT CREDENTIALS, based on the C2PA standard (Coalition for Content Provenance and Authenticity), attach signed information to a file, such as which camera or tool made it and what edits were applied. WATERMARKS are hidden or visible marks some generators add to their output. These help, but they have LIMITS. Records can be stripped by screenshots or re-uploading. A MISSING label does not prove content is real or fake. DETECTOR tools give scores, often shown as percentages, and make errors both ways.

A news photo shows Content Credentials listing the camera and edits. What does this give you?

A. Useful information about its history

B. A copyright license

C. Nothing useful

D. Absolute proof that everything shown in the photo is true, so no other checks are needed

REFERENCE LABEL: NOT PART OF THE QUESTION ID S84-F1-Q1
Artificial Intelligence, Set 84: Provenance labels and the limits of detection · FORWARD 1: APPLY IT · Q1/5 · Outcome AI.MEDIA.07 (local)

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A viral image has no credentials because it was screenshotted. What should you conclude?

A. That it is fake, because only fake images lose their credentials when they are shared

B. That it is old

C. That it is real

D. Nothing yet; check other clues

REFERENCE LABEL: NOT PART OF THE QUESTION ID S84-F1-Q2
Artificial Intelligence, Set 84: Provenance labels and the limits of detection · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.MEDIA.07 (local)

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A detector says a student essay's photo is "70% likely AI." What should a teacher do?

A. Ignore the photo
B. Treat it as one clue and talk with the student
C. Report the student publicly
D. Give the student a failing grade right away

REFERENCE LABEL: NOT PART OF THE QUESTION ID S84-F1-Q3
Artificial Intelligence, Set 84: Provenance labels and the limits of detection · FORWARD 1: APPLY IT · Q3/5 · Outcome AI.MEDIA.07 (local)

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A generator adds a visible watermark. What can remove it?

A. Viewing the image

B. Nothing at all, because watermarks are permanent and cannot be removed by any method

C. Cropping or editing

D. Saving the file

REFERENCE LABEL: NOT PART OF THE QUESTION ID S84-F1-Q4
Artificial Intelligence, Set 84: Provenance labels and the limits of detection · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.MEDIA.07 (local)

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Which approach to a suspicious image is best?

A. Rely only on how it looks

B. Rely only on likes

C. Rely only on a detector score, since detectors are designed to replace every other check

D. Combine provenance with source and context checks

REFERENCE LABEL: NOT PART OF THE QUESTION ID S84-F1-Q5
Artificial Intelligence, Set 84: Provenance labels and the limits of detection · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.MEDIA.07 (local)

Learn
Detector scores are especially easy to misuse. A detector that says "85% likely AI" is not proof; it can wrongly flag real photos (FALSE POSITIVES) and miss fakes (FALSE NEGATIVES). Using a score alone to accuse someone, such as a student of cheating or a journalist of faking, can cause real harm. Treat provenance and detection as one clue among several, alongside source, context and confirmation.

A detector answers YES when it judges an image AI-made. What is a false positive?

A. Flagging a real photo as AI
B. Correctly passing a real photo
C. Correctly flagging an AI image
D. Missing an AI image

REFERENCE LABEL: NOT PART OF THE QUESTION ID S84-F2-Q1
Artificial Intelligence, Set 84: Provenance labels and the limits of detection · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.MEDIA.07 (local)

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For the same kind of detector (YES means "AI-made"), what is a false negative?

A. Correctly passing a real photo
B. Correctly finding an AI image
C. Missing an AI image
D. Flagging a real photo as AI

REFERENCE LABEL: NOT PART OF THE QUESTION ID S84-F2-Q2
Artificial Intelligence, Set 84: Provenance labels and the limits of detection · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.MEDIA.07 (local)

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Why is accusing someone based on a score alone harmful?

A. Because detector scores are always exactly 50%, so they never give any information

B. Because scores are secret

C. Because accusations are always illegal

D. Scores can be wrong

REFERENCE LABEL: NOT PART OF THE QUESTION ID S84-F2-Q3
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A journalist's real photo is flagged "90% AI." What should an editor do?

A. Retract the photo at once, since 90% is high enough to treat the score as certain proof

B. Check the source and camera files

C. Fire the journalist

D. Publish the score instead

REFERENCE LABEL: NOT PART OF THE QUESTION ID S84-F2-Q4
Artificial Intelligence, Set 84: Provenance labels and the limits of detection · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.MEDIA.07 (local)

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How should provenance and detection be treated?

A. As one clue among several
B. As useless
C. As legal proof
D. As the final answer

REFERENCE LABEL: NOT PART OF THE QUESTION ID S84-F2-Q5
Artificial Intelligence, Set 84: Provenance labels and the limits of detection · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.MEDIA.07 (local)

Set 85: Reading a tiny program

Outcome AI.CODE.01: Read a few lines of Python with defined terms and predict what they do

Module: Coding: code assistance

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn

A PROGRAM is a list of instructions a computer follows in order, top to bottom. In the language PYTHON, a VARIABLE is a name that stores a value: "price = 4" stores 4 under the name price (the "=" means "store," not "is equal"). "*" multiplies, "+" adds and "/" divides. print(...) shows a value on screen. To read code, go line by line and track each variable's value, like keeping score on paper. In the example, price is 4 and count is 3, so total is 4 * 3 = 12, and the last line prints 12.

Prepared code example (Python 3). Shown only; nothing runs.

```
price = 4
count = 3
total = price * count
print(total)
```

In this example, what does the program print?

A. 7, because the program adds the price and the count together before printing the total

B. total

C. 12

D. 4

REFERENCE LABEL: NOT PART OF THE QUESTION ID S85-M-Q1

Artificial Intelligence, Set 85: Reading a tiny program · MAIN · Q1/5 · Outcome AI.CODE.01 (local)

Learn
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What does "price = 4" do in Python?

A. Stores 4 under the name price

B. Deletes price

C. Checks whether price is already equal to 4 and prints True or False on the screen

D. Prints 4

REFERENCE LABEL: NOT PART OF THE QUESTION ID S85-M-Q2
Artificial Intelligence, Set 85: Reading a tiny program · MAIN · Q2/5 · Outcome AI.CODE.01 (local)

Learn

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Prepared code example (Python 3). Shown only; nothing runs.

```
price = 4
count = 3
total = price * count
print(total)
```

If the second line were "count = 5," what would print?

A. 9, because changing the count to 5 adds 5 to the price of 4 instead of multiplying

B. 5

C. 12

D. 20

REFERENCE LABEL: NOT PART OF THE QUESTION ID S85-M-Q3
Artificial Intelligence, Set 85: Reading a tiny program · MAIN · Q3/5 · Outcome AI.CODE.01 (local)

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In what order does Python run these lines?

A. In random order
B. Top to bottom
C. Only the last line
D. Bottom to top

REFERENCE LABEL: NOT PART OF THE QUESTION ID S85-M-Q4
Artificial Intelligence, Set 85: Reading a tiny program · MAIN · Q4/5 · Outcome AI.CODE.01 (local)

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What does print(...) do?

A. Multiplies numbers
B. Stores a value
C. Shows a value on screen
D. Sends the program

REFERENCE LABEL: NOT PART OF THE QUESTION ID S85-M-Q5
Artificial Intelligence, Set 85: Reading a tiny program · MAIN · Q5/5 · Outcome AI.CODE.01 (local)

Learn
TRACING means following a program line by line and writing down each variable's value as it changes, like a scoreboard. Example: line 1 "a = 2" (a is 2). Line 2 "b = a + 3" (b is 5). Line 3 "a = 10" (a is now 10; b stays 5, because b was already calculated). Line 4 "print(a + b)" shows 15. The key habit: when a variable changes, cross out its old value and write the new one, and do not change other variables unless a line says so.

After line 2, what is b?

A. 5

B. 3

C. 10

D. 2, because b is given the same value as a, and the "+ 3" part is ignored by Python

REFERENCE LABEL: NOT PART OF THE QUESTION ID S85-B1-Q1
Artificial Intelligence, Set 85: Reading a tiny program · BACKWARD 1 (underlying skill) · Q1/5 · Outcome AI.CODE.01 (local)

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After line 3, what is a?

A. 12

B. 10

C. 2, because once a variable is given a value it can never be changed by a later line

D. 5

REFERENCE LABEL: NOT PART OF THE QUESTION ID S85-B1-Q2
Artificial Intelligence, Set 85: Reading a tiny program · BACKWARD 1 (underlying skill) · Q2/5 · Outcome AI.CODE.01 (local)

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After line 3, what is b?

A. 10
B. 2
C. 13
D. 5

REFERENCE LABEL: NOT PART OF THE QUESTION ID S85-B1-Q3
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What does line 4 print?

A. 12

B. 7, which is the original value of a added to the original value of b from line 2

C. 15

D. a + b

REFERENCE LABEL: NOT PART OF THE QUESTION ID S85-B1-Q4
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What is the key tracing habit?

A. Track only the last line

B. Update every variable on every line, even when the line does not mention that variable

C. Ignore numbers

D. Update a variable only when a line changes it

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Prepared code example (Python 3). Shown only; nothing runs.

```
age = 15
if age >= 16:
    print("Can apply")
else:
    print("Too young")
```

What does this program print?

A. 15
B. Too young
C. Nothing
D. Can apply

REFERENCE LABEL: NOT PART OF THE QUESTION ID S85-F1-Q1
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```
age = 15
if age >= 16:
    print("Can apply")
else:
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```

If the first line were "age = 16," what would print?

A. Can apply
B. 16
C. Both lines
D. Too young

REFERENCE LABEL: NOT PART OF THE QUESTION ID S85-F1-Q2
Artificial Intelligence, Set 85: Reading a tiny program · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.CODE.01 (local)

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Prepared code example (Python 3). Shown only; nothing runs.

```
for n in [1, 2, 3]:
    print(n * 2)
```

What does this loop print?

A. 12

B. 6 only

C. 1, 2 and 3 on separate lines

D. 2, 4 and 6 on separate lines

REFERENCE LABEL: NOT PART OF THE QUESTION ID S85-F1-Q3

Artificial Intelligence, Set 85: Reading a tiny program · FORWARD 1: APPLY IT · Q3/5 · Outcome AI.CODE.01 (local)

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Prepared code example (Python 3). Shown only; nothing runs.

```
for n in [1, 2, 3]:
    print(n * 2)
```

How many times does the loop body run?

A. 6, because the loop runs once for every number that it prints out on the screen

B. 3

C. 2

D. 1

REFERENCE LABEL: NOT PART OF THE QUESTION ID S85-F1-Q4

Artificial Intelligence, Set 85: Reading a tiny program · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.CODE.01 (local)

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In "x = 7" then "x = x + 1," what is x afterward?

A. 8

B. 7, because a variable cannot be used on both sides of the same "=" sign in Python

C. 71

D. 1

REFERENCE LABEL: NOT PART OF THE QUESTION ID S85-F1-Q5
Artificial Intelligence, Set 85: Reading a tiny program · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.CODE.01 (local)

Learn
Two more patterns appear constantly. An IF statement runs lines only when a condition is true: "if age >= 16:" means "if age is at least 16." Lines indented under it belong to it, and an ELSE part runs otherwise. A FOR LOOP repeats lines once for each item in a list: "for n in [1, 2, 3]:" runs its indented lines three times, with n being 1, then 2, then 3. Indentation (spaces at the start of a line) shows which lines belong together.

What does "if age >= 16:" mean?

A. Set age to 16
B. If age is under 16
C. If age is at least 16
D. If age is exactly 16

REFERENCE LABEL: NOT PART OF THE QUESTION ID S85-F2-Q1
Artificial Intelligence, Set 85: Reading a tiny program · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.CODE.01 (local)

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What does indentation show in Python?

A. Which lines belong together
B. Comments
C. How important each line is, with the most important lines indented the furthest
D. Nothing; it is decoration

REFERENCE LABEL: NOT PART OF THE QUESTION ID S85-F2-Q2
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In "for n in [5, 10]: print(n)," what prints?

A. Nothing
B. n twice
C. 5 then 10
D. 15

REFERENCE LABEL: NOT PART OF THE QUESTION ID S85-F2-Q3
Artificial Intelligence, Set 85: Reading a tiny program · FORWARD 2: HANDLE THE COMPLICATIONS · Q3/5 · Outcome AI.CODE.01 (local)

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When does an else part run?

A. Never

B. Always, right after the if part has finished, whether the condition was true or false

C. Before the if

D. When the if condition is false

REFERENCE LABEL: NOT PART OF THE QUESTION ID S85-F2-Q4
Artificial Intelligence, Set 85: Reading a tiny program · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.CODE.01 (local)

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A loop goes over a list of 4 names. How many times does its body run?

A. 8

B. 4

C. 0

D. 1, because a loop body runs only once no matter how many items are in the list

REFERENCE LABEL: NOT PART OF THE QUESTION ID S85-F2-Q5
Artificial Intelligence, Set 85: Reading a tiny program · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.CODE.01 (local)

Set 86: Why generated code needs checking

Outcome AI.CODE.02: Explain why code that looks right can still be wrong and how to check it

Module: Coding: code assistance

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
AI-generated code often LOOKS right: neat layout, sensible names, helpful comments. It can still be WRONG, because the model predicts likely code rather than proving it correct. Common problems: logic that works for one example but not others, misunderstood requirements, missing cases such as an empty list, and calls to functions or settings that do not exist. To check code: read it and trace it, RUN it on several inputs whose correct answers you already know, and compare results. Passing one test does not prove the code is right.

Why can generated code be wrong even when it looks neat?

A. The model predicts likely code without proving it

B. Because computers cannot run neat code

C. Because comments stop code from running

D. Because neat code is always wrong, and only messy code can actually work correctly

REFERENCE LABEL: NOT PART OF THE QUESTION ID S86-M-Q1
Artificial Intelligence, Set 86: Why generated code needs checking · MAIN · Q1/5 · Outcome AI.CODE.02 (local)

Learn

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Prepared code example (Python 3). Shown only; nothing runs.

```
def average(numbers):
    return sum(numbers) / 2

print(average([4, 6]))
print(average([3, 6, 9]))
```

In this example, which result is wrong?

A. Both results

B. Neither result

C. 5.0 for [4, 6]

D. 9.0 for [3, 6, 9]

REFERENCE LABEL: NOT PART OF THE QUESTION ID S86-M-Q2

Artificial Intelligence, Set 86: Why generated code needs checking · MAIN · Q2/5 · Outcome AI.CODE.02 (local)

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```
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```

What is the bug in the same example?

A. Its name is wrong
B. It prints too much
C. It always divides by 2
D. It uses sum

REFERENCE LABEL: NOT PART OF THE QUESTION ID S86-M-Q3
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```
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print(average([4, 6]))
print(average([3, 6, 9]))
```

What does the example show about testing?

A. Code never needs tests

B. Passing one test does not prove code right

C. Testing is a waste of time, because the first test already showed that the function works

D. Two tests are always enough

REFERENCE LABEL: NOT PART OF THE QUESTION ID S86-M-Q4
Artificial Intelligence, Set 86: Why generated code needs checking · MAIN · Q4/5 · Outcome AI.CODE.02 (local)

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What is the best way to check code?

A. Count its lines

B. Ask the tool that wrote it whether it is correct, and accept the answer that it gives

C. Run it on inputs whose answers you know

D. Check that it has comments

REFERENCE LABEL: NOT PART OF THE QUESTION ID S86-M-Q5
Artificial Intelligence, Set 86: Why generated code needs checking · MAIN · Q5/5 · Outcome AI.CODE.02 (local)

Learn
Imagine you built a calculator and want to know if it works. You would not just admire the buttons. You would type problems you already know the answer to: 2 + 2, 10 - 3, 5 * 5. If one gives a wrong result, you know something is broken, even if the others were right. Generated code is like a new calculator: neat appearance proves nothing; known-answer tests do.

Why type 2 + 2 into a new calculator?

A. To make it faster
B. You already know the answer
C. To clean the buttons
D. Because 2 + 2 is the only problem

REFERENCE LABEL: NOT PART OF THE QUESTION ID S86-B1-Q1
Artificial Intelligence, Set 86: Why generated code needs checking · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.CODE.02 (local)

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The calculator gets 2 + 2 right but 5 * 5 wrong. Does it work?

A. No; one wrong result shows a problem

B. It cannot be known

C. Yes, mostly

D. Yes, because getting the first problem right proves that every other answer is right

REFERENCE LABEL: NOT PART OF THE QUESTION ID S86-B1-Q2
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Nice-looking buttons prove what?

A. That every answer the calculator gives will always be correct for any problem
B. That it is cheap
C. That it is fast
D. Nothing about correctness

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Generated code is like what in this comparison?

A. A broken calculator

B. A new calculator to test

C. A math book

D. A calculator that has already been fully tested by thousands of people over many years

REFERENCE LABEL: NOT PART OF THE QUESTION ID S86-B1-Q4
Artificial Intelligence, Set 86: Why generated code needs checking · BACKWARD 1 (concept simplification) · Q4/5 · Outcome AI.CODE.02 (local)

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What should you test code with?

A. Inputs chosen at random whose correct answers nobody knows, so the test is fair

B. Only one input

C. No inputs

D. Inputs whose correct answers you know

REFERENCE LABEL: NOT PART OF THE QUESTION ID S86-B1-Q5
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A tool writes a function to count vowels. Which test helps most?

A. Check it has a good name

B. Count its lines

C. Run it on "banana" and expect 3

D. Read the comment at the top

REFERENCE LABEL: NOT PART OF THE QUESTION ID S86-F1-Q1

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Generated code calls a function the tool named "sort_smart," which does not exist. What will happen?

A. An error when it runs

B. The computer will create the missing function automatically, based on its name

C. It will run normally

D. It will print the name

REFERENCE LABEL: NOT PART OF THE QUESTION ID S86-F1-Q2
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A function to find the largest number works on [3, 9, 2]. What else should you test?

A. The same list again

B. An empty list and negative numbers

C. Nothing else, since working on one list of three numbers shows it will work on every list

D. A list of colors

REFERENCE LABEL: NOT PART OF THE QUESTION ID S86-F1-Q3
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Code will calculate students' grades. What extra step is wise?

A. Have someone experienced review it
B. Make it shorter
C. Delete the tests
D. Use it immediately for every student

REFERENCE LABEL: NOT PART OF THE QUESTION ID S86-F1-Q4
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Which shows that code really works?

A. Neat indentation

B. A long, confident explanation from the tool describing exactly what the code is doing

C. Correct results on several known-answer tests

D. Many comments

REFERENCE LABEL: NOT PART OF THE QUESTION ID S86-F1-Q5
Artificial Intelligence, Set 86: Why generated code needs checking · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.CODE.02 (local)

Learn
Confident explanations make wrong code more convincing. A tool may describe what code "does" based on what it was supposed to do, not what it actually does. Comments can be wrong too. Trust behavior over descriptions: run it, and test the boundaries (the smallest, largest, empty and unusual inputs). When the code will affect money, safety, privacy or other people, have someone experienced review it before it is used.

A tool explains its code "removes duplicates," but running it leaves duplicates. Which should you trust?

A. Neither

B. The explanation, since the tool knows its own code better than any test that you could run

C. The comments

D. What the code actually does when run

REFERENCE LABEL: NOT PART OF THE QUESTION ID S86-F2-Q1
Artificial Intelligence, Set 86: Why generated code needs checking · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.CODE.02 (local)

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A comment says "# adds tax," but the line subtracts. What does this show?

A. Comments can be wrong

B. That the subtraction must be correct, because comments are always checked before code runs

C. That the code will not run

D. That tax is negative

REFERENCE LABEL: NOT PART OF THE QUESTION ID S86-F2-Q2
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Which inputs test the boundaries of a function that handles ages?

A. Only 25

B. Only names

C. Three typical adult ages, such as 30, 35 and 40, which cover the most common situations

D. 0, a very large age, and a missing age

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Why do confident explanations make wrong code more convincing?

A. Because they contain numbers

B. They make you less likely to test it

C. Because confident explanations actually change the code so that it works correctly

D. Because they are longer

REFERENCE LABEL: NOT PART OF THE QUESTION ID S86-F2-Q4
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Code will handle customer payments. What should happen before use?

A. Only a spell check

B. Only a quick glance

C. Experienced review and testing

D. Nothing, since payment code written by a tool is always reviewed by a bank automatically

REFERENCE LABEL: NOT PART OF THE QUESTION ID S86-F2-Q5
Artificial Intelligence, Set 86: Why generated code needs checking · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.CODE.02 (local)

Set 87: Requirements and edge cases

Outcome AI.CODE.03: State requirements clearly, including unusual inputs (edge cases)

Module: Coding: code assistance

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A REQUIREMENT says what the code must do, stated clearly enough to check: "Given a list of test scores from 0 to 100, return the average rounded to one decimal place." EDGE CASES are unusual inputs that often break code: an EMPTY input (no scores), ZERO, NEGATIVE numbers, very LARGE values, MISSING or WRONG-TYPE data (a score typed as "ninety"), and BOUNDARY values exactly at a limit (0 and 100). When you ask a tool for code, state the requirement and the edge cases, and say what should happen for each, such as "if the list is empty, return 0 and do not crash."

What is a requirement?

A. A guess about what users want
B. The program's file name
C. A checkable statement of what code must do
D. A list of the programming languages

REFERENCE LABEL: NOT PART OF THE QUESTION ID S87-M-Q1
Artificial Intelligence, Set 87: Requirements and edge cases · MAIN · Q1/5 · Outcome AI.CODE.03 (local)

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Which is an edge case for a test-score average?

A. An empty list of scores
B. A list of ten scores
C. Scores of 80 and 85
D. A typical list

REFERENCE LABEL: NOT PART OF THE QUESTION ID S87-M-Q2
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Prepared code example (Python 3). Shown only; nothing runs.

```
def first_letter(name):
    return name[0]

print(first_letter("Maya"))
```

In this example, which input would break the function?

A. "Zoe"

B. An empty name ""

C. "Al"

D. "Maya," since names with four letters are too long for this function to handle correctly

REFERENCE LABEL: NOT PART OF THE QUESTION ID S87-M-Q3
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Which is a boundary value for scores from 0 to 100?

A. 72
B. 85
C. 57
D. 100

REFERENCE LABEL: NOT PART OF THE QUESTION ID S87-M-Q4
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Which request includes an edge-case rule?

A. "If the list is empty, return 0 and do not crash."

B. "Use good names."

C. "Make it fast."

D. "Write the code for the average quickly, and make it look clean and professional."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S87-M-Q5
Artificial Intelligence, Set 87: Requirements and edge cases · MAIN · Q5/5 · Outcome AI.CODE.03 (local)

Learn
When a locksmith tests a new lock, they do not only try the right key once. They try the wrong key, no key, a key pushed in halfway, turning it very hard, and the lock in freezing weather. These UNUSUAL situations are where locks fail. Edge cases in code are the same idea: the unusual inputs, such as empty, zero, negative or wrong-type, where code is most likely to break.

Why does a locksmith try the wrong key?

A. To save time
B. To see how the lock handles unusual situations
C. To damage the lock
D. Because the right key never works on any new lock

REFERENCE LABEL: NOT PART OF THE QUESTION ID S87-B1-Q1
Artificial Intelligence, Set 87: Requirements and edge cases · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.CODE.03 (local)

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"No key at all" is like which code input?

A. A very large number
B. A correct name
C. An empty input
D. A typical input

REFERENCE LABEL: NOT PART OF THE QUESTION ID S87-B1-Q2
Artificial Intelligence, Set 87: Requirements and edge cases · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.CODE.03 (local)

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Freezing weather is like what in code?

A. A comment
B. A variable name
C. Normal use
D. Unusual conditions

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Where do locks and code most often fail?

A. In unusual situations
B. Never
C. Only on weekends
D. In the most common situation, the one that the designer tested most carefully

REFERENCE LABEL: NOT PART OF THE QUESTION ID S87-B1-Q4
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What is the lesson for code?

A. Only test long inputs

B. Never test code

C. Test only the normal input once, because unusual inputs never happen in real life

D. Test unusual inputs, not just the normal one

REFERENCE LABEL: NOT PART OF THE QUESTION ID S87-B1-Q5
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Ben asks: "Write code to split a bill." What key detail is missing?

A. The restaurant's address
B. The programmer's name
C. How many people and how to round
D. Which color to use for the numbers

REFERENCE LABEL: NOT PART OF THE QUESTION ID S87-F1-Q1
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A function divides a total by the number of people. Which edge case matters most?

A. Three people
B. Zero people
C. Two people
D. Four people

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Which requirement is clearest?

A. "Do prices."
B. "Handle tax."
C. "Make the price thing work properly with tax and everything else."
D. "Return the total price including 8% tax, rounded to cents."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S87-F1-Q3
Artificial Intelligence, Set 87: Requirements and edge cases · FORWARD 1: APPLY IT · Q3/5 · Outcome AI.CODE.03 (local)

Learn
A REQUIREMENT says what the code must do, stated clearly enough to check: "Given a list of test scores from 0 to 100, return the average rounded to one decimal place." EDGE CASES are unusual inputs that often break code: an EMPTY input (no scores), ZERO, NEGATIVE numbers, very LARGE values, MISSING or WRONG-TYPE data (a score typed as "ninety"), and BOUNDARY values exactly at a limit (0 and 100). When you ask a tool for code, state the requirement and the edge cases, and say what should happen for each, such as "if the list is empty, return 0 and do not crash."

A form asks for age. Which input is a wrong-type edge case?

A. 15

B. "twelve" typed as a word

C. 30

D. 12, typed as a whole number

REFERENCE LABEL: NOT PART OF THE QUESTION ID S87-F1-Q4
Artificial Intelligence, Set 87: Requirements and edge cases · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.CODE.03 (local)

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A score must be from 0 to 100. Which two inputs test the boundaries?

A. 60 and 61
B. 20 and 30
C. 0 and 100
D. 50 and 75

REFERENCE LABEL: NOT PART OF THE QUESTION ID S87-F1-Q5
Artificial Intelligence, Set 87: Requirements and edge cases · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.CODE.03 (local)

Learn
Unstated requirements get filled in by guesses. "Sort the names" leaves open: alphabetical or by length? Uppercase before lowercase? What about "Ana" and "ana"? A tool will pick something, and it may not be what you need. Write down examples of inputs with the expected outputs; these EXAMPLES double as tests later. If you are not sure what should happen in a case, decide that first, because the code cannot be right about a rule nobody chose.

"Sort the names" leaves which question open?

A. Alphabetical or by length?
B. Which font to show
C. Which keyboard to use
D. Nothing

REFERENCE LABEL: NOT PART OF THE QUESTION ID S87-F2-Q1
Artificial Intelligence, Set 87: Requirements and edge cases · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.CODE.03 (local)

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Why write input examples with expected outputs?

A. Because tools need exactly ten examples
B. They clarify the requirement and become tests
C. To make the request longer
D. Because examples make the code run faster

REFERENCE LABEL: NOT PART OF THE QUESTION ID S87-F2-Q2
Artificial Intelligence, Set 87: Requirements and edge cases · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.CODE.03 (local)

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You are unsure whether "Ana" and "ana" count as the same name. What should you do?

A. Delete one of the names
B. Ignore the question
C. Let the tool decide quietly
D. Decide the rule before coding

REFERENCE LABEL: NOT PART OF THE QUESTION ID S87-F2-Q3
Artificial Intelligence, Set 87: Requirements and edge cases · FORWARD 2: HANDLE THE COMPLICATIONS · Q3/5 · Outcome AI.CODE.03 (local)

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What happens when a requirement is unstated?

A. The tool fills it with a guess

B. The tool always asks you a question first and never guesses anything on its own

C. The code refuses to run

D. Nothing at all

REFERENCE LABEL: NOT PART OF THE QUESTION ID S87-F2-Q4
Artificial Intelligence, Set 87: Requirements and edge cases · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.CODE.03 (local)

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Which example helps clarify "remove extra spaces"?

A. "hello" stays "hello"

B. "Spaces are bad"

C. " hi there " becomes "hi there"

D. "Make the text look tidier and cleaner"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S87-F2-Q5
Artificial Intelligence, Set 87: Requirements and edge cases · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.CODE.03 (local)

Set 88: Tests that catch mistakes

Outcome AI.CODE.04: Choose a test whose expected result would reveal a particular defect

Module: Coding: code assistance

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
A TEST runs code on a chosen input and compares the result with the EXPECTED result you worked out yourself. A good test is one that would FAIL if a particular mistake were present. To catch a mistake at a limit, test the limit itself. Example: the rule is "18 or older is an adult." A function using "age > 18" looks right and gives True for 20, but it wrongly gives False for 18. Only a test with age 18 reveals this. Choose tests by asking: "What mistake might be here, and which input would expose it?"

What does a test compare?

A. The code's speed with the internet speed
B. Two programmers' opinions
C. The code's length with the length of another program
D. The code's result with an expected result you worked out

REFERENCE LABEL: NOT PART OF THE QUESTION ID S88-M-Q1
Artificial Intelligence, Set 88: Tests that catch mistakes · MAIN · Q1/5 · Outcome AI.CODE.04 (local)

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Prepared code example (Python 3). Shown only; nothing runs.

```
def is_adult(age):
    return age > 18

print(is_adult(20))
print(is_adult(18))
```

In this example, which test reveals the bug?

A. is_adult(18)
B. is_adult(50)
C. is_adult(30)
D. is_adult(20)

REFERENCE LABEL: NOT PART OF THE QUESTION ID S88-M-Q2
Artificial Intelligence, Set 88: Tests that catch mistakes · MAIN · Q2/5 · Outcome AI.CODE.04 (local)

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Prepared code example (Python 3). Shown only; nothing runs.

```
def is_adult(age):
    return age > 18

print(is_adult(20))
print(is_adult(18))
```

What is the bug in the same example?

A. It prints too much
B. It uses > instead of >=
C. It has no name
D. It uses the word "age,"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S88-M-Q3
Artificial Intelligence, Set 88: Tests that catch mistakes · MAIN · Q3/5 · Outcome AI.CODE.04 (local)

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What makes a test good?

A. It is very long

B. It uses random inputs

C. It would fail if a particular mistake were present

D. It always passes, which shows that the code is working correctly every single time

REFERENCE LABEL: NOT PART OF THE QUESTION ID S88-M-Q4
Artificial Intelligence, Set 88: Tests that catch mistakes · MAIN · Q4/5 · Outcome AI.CODE.04 (local)

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To catch a mistake at a limit, what should you test?

A. Nothing
B. The limit itself
C. Only typical values
D. A value far away

REFERENCE LABEL: NOT PART OF THE QUESTION ID S88-M-Q5
Artificial Intelligence, Set 88: Tests that catch mistakes · MAIN · Q5/5 · Outcome AI.CODE.04 (local)

Learn
Work backward from the mistake. Ask: "If the code had this mistake, which input would give a different answer from the correct code?" Example: a free-shipping rule says "orders of 50 dollars or more ship free." Suspected mistake: using > 50 instead of >= 50. Inputs 30 and 80 give the same answers either way. Only input 50 differs: the correct rule says free, the mistaken one says not free. So 50 is the test to run, expecting "free."

Which input exposes > 50 versus >= 50?

A. 30
B. 100
C. 50
D. 80

REFERENCE LABEL: NOT PART OF THE QUESTION ID S88-B1-Q1
Artificial Intelligence, Set 88: Tests that catch mistakes · BACKWARD 1 (underlying skill) · Q1/5 · Outcome AI.CODE.04 (local)

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For input 50, what should the test expect?

A. Free shipping
B. An error
C. Half-price shipping
D. Not free

REFERENCE LABEL: NOT PART OF THE QUESTION ID S88-B1-Q2
Artificial Intelligence, Set 88: Tests that catch mistakes · BACKWARD 1 (underlying skill) · Q2/5 · Outcome AI.CODE.04 (local)

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Why do 30 and 80 not help?

A. They crash the code
B. They are too large
C. Because 30 and 80 are not real prices
D. Both rules give the same answer for them

REFERENCE LABEL: NOT PART OF THE QUESTION ID S88-B1-Q3
Artificial Intelligence, Set 88: Tests that catch mistakes · BACKWARD 1 (underlying skill) · Q3/5 · Outcome AI.CODE.04 (local)

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What question drives the method?

A. Which input is easiest to type?

B. Which input is the most common?

C. Which input gives a different answer if the mistake exists?

D. Which input is the largest number that the program is able to handle without crashing?

REFERENCE LABEL: NOT PART OF THE QUESTION ID S88-B1-Q4
Artificial Intelligence, Set 88: Tests that catch mistakes · BACKWARD 1 (underlying skill) · Q4/5 · Outcome AI.CODE.04 (local)

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Work backward from the mistake. Ask: "If the code had this mistake, which input would give a different answer from the correct code?" Example: a free-shipping rule says "orders of 50 dollars or more ship free." Suspected mistake: using > 50 instead of >= 50. Inputs 30 and 80 give the same answers either way. Only input 50 differs: the correct rule says free, the mistaken one says not free. So 50 is the test to run, expecting "free."

Rule: "under 13 pays child price." Suspected mistake: <= 13. Which input exposes it?

A. 12

B. 5, because young children are the main customers who pay the child price at the door

C. 20

D. 13

REFERENCE LABEL: NOT PART OF THE QUESTION ID S88-B1-Q5
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A function should give a discount to orders of 100 dollars or more. Which test is most useful?

A. An order of exactly 100, expecting a discount

B. An order of 500, which is so large that it should definitely get the discount in any case

C. No test

D. An order of 20, expecting no discount

REFERENCE LABEL: NOT PART OF THE QUESTION ID S88-F1-Q1
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A function should count words. Which test could reveal a mistake with extra spaces?

A. "hello," a single word with no spaces at all, which is the simplest possible test
B. "hi there" (two spaces), expecting 2
C. An empty test
D. "one two" with one space

REFERENCE LABEL: NOT PART OF THE QUESTION ID S88-F1-Q2
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Ana writes a test expecting the average of [2, 4] to be 3. The code returns 3. What does this show?

A. That the test is wrong
B. Nothing at all
C. That the code is correct
D. The code passes this test

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A function should reject passwords shorter than 8 characters. Which input tests the boundary?

A. A password of 30 characters, since longer passwords are the most important ones to check

B. A password of exactly 7 characters, expecting rejection

C. A password of 15 characters

D. No password

REFERENCE LABEL: NOT PART OF THE QUESTION ID S88-F1-Q4
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Which expected result should a test use?

A. The tool's guess

B. Whatever the code printed the first time it ran, since the code is the best source of answers

C. One you worked out yourself

D. A random number

REFERENCE LABEL: NOT PART OF THE QUESTION ID S88-F1-Q5
Artificial Intelligence, Set 88: Tests that catch mistakes · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.CODE.04 (local)

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Tests can be weak in several ways. A test whose expected answer was copied from the code's own output only checks that the code agrees with itself. A test that uses only typical inputs may never reach the buggy part. And a test that checks too little, such as "the result is a number," passes even when the number is wrong. Strong tests use expected answers worked out independently and target the boundaries and special cases.

A test's expected answer was copied from the code's output. What is the problem?

A. It only checks the code agrees with itself

B. The test uses numbers

C. Nothing, because the code's own output is always the most reliable expected answer

D. The test is too long

REFERENCE LABEL: NOT PART OF THE QUESTION ID S88-F2-Q1
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A test checks only that the result "is a number." Why is it weak?

A. Because numbers are not results
B. Because it is too strict
C. It passes even when the number is wrong
D. Because numbers can never be checked by tests

REFERENCE LABEL: NOT PART OF THE QUESTION ID S88-F2-Q2
Artificial Intelligence, Set 88: Tests that catch mistakes · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.CODE.04 (local)

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All tests use typical inputs. What might they miss?

A. Spelling mistakes in comments

B. Bugs at boundaries and special cases

C. Nothing, since typical inputs exercise every single line of code in every program

D. The program's file name

REFERENCE LABEL: NOT PART OF THE QUESTION ID S88-F2-Q3
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Which test is strongest for a function that returns the largest of three numbers?

A. No test
B. (5, 5, 5) expecting a result
C. (9, 2, 1) expecting a number of some kind
D. Largest in last place: (1, 2, 9) expecting 9

REFERENCE LABEL: NOT PART OF THE QUESTION ID S88-F2-Q4
Artificial Intelligence, Set 88: Tests that catch mistakes · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.CODE.04 (local)

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Where should strong tests focus?

A. Boundaries and special cases, with independent answers
B. Only the program name
C. Random inputs with no expected answers
D. The most common input only, run many times in a row

REFERENCE LABEL: NOT PART OF THE QUESTION ID S88-F2-Q5
Artificial Intelligence, Set 88: Tests that catch mistakes · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.CODE.04 (local)

Set 89: Reading error messages

Outcome AI.CODE.05: Use an error message to locate a problem before asking for a fix

Module: Coding: code assistance

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
When Python hits a problem, it stops and shows an ERROR MESSAGE, often called a TRACEBACK. Read it from the BOTTOM: the last line names the error type and a short explanation, and the line above names the FILE and LINE NUMBER where it happened. Common types: NameError (a name that was never defined, often a typo), TypeError (mixing types, such as adding text to a number), IndexError (asking for a position that does not exist) and ValueError (a value of the right type but wrong content). Use the message to find the problem before asking a tool to fix it, and include the relevant message, with any keys or personal details removed, when you ask.

Where should you start reading a traceback?

A. At the bottom
B. In the middle
C. Nowhere; ignore it
D. At the very top

REFERENCE LABEL: NOT PART OF THE QUESTION ID S89-M-Q1
Artificial Intelligence, Set 89: Reading error messages · MAIN · Q1/5 · Outcome AI.CODE.05 (local)

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Prepared code example (Python 3) and its error message, simplified. Shown only; nothing runs.

```
scores = [90, 85, 77]
print(scores[3])

Traceback (most recent call last):
File "grades.py", line 2, in <module>
print(scores[3])
IndexError: list index out of range
```

In this example, what type of error is shown?

A. TypeError
B. IndexError
C. ValueError
D. NameError

REFERENCE LABEL: NOT PART OF THE QUESTION ID S89-M-Q2
Artificial Intelligence, Set 89: Reading error messages · MAIN · Q2/5 · Outcome AI.CODE.05 (local)

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scores = [90, 85, 77]
print(scores[3])

Traceback (most recent call last):
File "grades.py", line 2, in <module>
print(scores[3])
IndexError: list index out of range
```

In the same example, why is position 3 out of range?

A. Because 77 is too large

B. Because print cannot show lists

C. Python counts positions from 0, so they are 0 to 2

D. Because lists in Python can only hold two items

REFERENCE LABEL: NOT PART OF THE QUESTION ID S89-M-Q3
Artificial Intelligence, Set 89: Reading error messages · MAIN · Q3/5 · Outcome AI.CODE.05 (local)

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Prepared code example (Python 3) and its error message, simplified. Shown only; nothing runs.

```
total = 50
print(totl)

Traceback (most recent call last):
File "shop.py", line 2, in <module>
print(totl)
NameError: name 'totl' is not defined
```

In this example, what caused the NameError?

A. print is misspelled

B. The file name is wrong

C. The number 50 is too large

D. A typo: totl instead of total

REFERENCE LABEL: NOT PART OF THE QUESTION ID S89-M-Q4
Artificial Intelligence, Set 89: Reading error messages · MAIN · Q4/5 · Outcome AI.CODE.05 (local)

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When asking a tool to fix an error, what should you include?

A. Only the words "it is broken," since the tool can work out the rest by itself
B. The relevant error message, with secrets removed
C. Nothing
D. Only the file name

REFERENCE LABEL: NOT PART OF THE QUESTION ID S89-M-Q5
Artificial Intelligence, Set 89: Reading error messages · MAIN · Q5/5 · Outcome AI.CODE.05 (local)

Learn
Most of what you need is in the last two lines of a traceback. The LAST line has the error TYPE and a short explanation. The line just above it (usually) shows the code that failed, and the line above that names the FILE and LINE NUMBER. Example ending: 'File "pay.py", line 7' then 'rate = hours / 0' then 'ZeroDivisionError: division by zero.' That tells you: file pay.py, line 7, dividing by zero.

In the example, which file has the error?

A. line 7
B. ZeroDivisionError, which is the name of the file where the problem happened
C. pay.py
D. rate

REFERENCE LABEL: NOT PART OF THE QUESTION ID S89-B1-Q1
Artificial Intelligence, Set 89: Reading error messages · BACKWARD 1 (underlying skill) · Q1/5 · Outcome AI.CODE.05 (local)

Learn
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Which line number?

A. 7

B. 0, because the error message mentions division by zero in the last line

C. 1

D. 2

REFERENCE LABEL: NOT PART OF THE QUESTION ID S89-B1-Q2
Artificial Intelligence, Set 89: Reading error messages · BACKWARD 1 (underlying skill) · Q2/5 · Outcome AI.CODE.05 (local)

Learn
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What is the error type?

A. hours
B. line 7
C. pay.py, which is listed just above the line of code that caused the problem
D. ZeroDivisionError

REFERENCE LABEL: NOT PART OF THE QUESTION ID S89-B1-Q3
Artificial Intelligence, Set 89: Reading error messages · BACKWARD 1 (underlying skill) · Q3/5 · Outcome AI.CODE.05 (local)

Learn
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What is the problem?

A. Adding text to numbers
B. Dividing by zero
C. A missing file
D. A misspelled variable name

REFERENCE LABEL: NOT PART OF THE QUESTION ID S89-B1-Q4
Artificial Intelligence, Set 89: Reading error messages · BACKWARD 1 (underlying skill) · Q4/5 · Outcome AI.CODE.05 (local)

Learn
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Which two lines hold most of the information?

A. The last two
B. None
C. The middle two
D. The first two

REFERENCE LABEL: NOT PART OF THE QUESTION ID S89-B1-Q5
Artificial Intelligence, Set 89: Reading error messages · BACKWARD 1 (underlying skill) · Q5/5 · Outcome AI.CODE.05 (local)

Learn

When Python hits a problem, it stops and shows an ERROR MESSAGE, often called a TRACEBACK. Read it from the BOTTOM: the last line names the error type and a short explanation, and the line above names the FILE and LINE NUMBER where it happened. Common types: NameError (a name that was never defined, often a typo), TypeError (mixing types, such as adding text to a number), IndexError (asking for a position that does not exist) and ValueError (a value of the right type but wrong content). Use the message to find the problem before asking a tool to fix it, and include the relevant message, with any keys or personal details removed, when you ask.

Prepared code example (Python 3) and its error message, simplified. Shown only; nothing runs.

```
age = 15
print("Age: " + age)

Traceback (most recent call last):
File "card.py", line 2, in <module>
print("Age: " + age)
TypeError: can only concatenate str (not "int") to str
```

In this example, what is the problem?

A. The variable name "age" is not allowed in Python, so the program cannot run at all

B. A missing file

C. The number 15 is too large

D. Adding text to a number

REFERENCE LABEL: NOT PART OF THE QUESTION ID S89-F1-Q1
Artificial Intelligence, Set 89: Reading error messages · FORWARD 1: APPLY IT · Q1/5 · Outcome AI.CODE.05 (local)

Learn

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age = 15
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Traceback (most recent call last):
File "card.py", line 2, in <module>
print("Age: " + age)
TypeError: can only concatenate str (not "int") to str
```

In the same example, which line has the problem?

A. Line 3

B. Line 15

C. Line 2

D. Line 1

REFERENCE LABEL: NOT PART OF THE QUESTION ID S89-F1-Q2

Artificial Intelligence, Set 89: Reading error messages · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.CODE.05 (local)

Learn
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int("ten") gives "ValueError: invalid literal for int()." What does it mean?

A. The list is too short
B. The file is missing
C. The program forgot to define a variable called "ten" before using it on this line
D. The text is not a number Python can convert

REFERENCE LABEL: NOT PART OF THE QUESTION ID S89-F1-Q3
Artificial Intelligence, Set 89: Reading error messages · FORWARD 1: APPLY IT · Q3/5 · Outcome AI.CODE.05 (local)

Learn
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A NameError names "prnt." What is the likely cause?

A. Text plus a number
B. Dividing by zero
C. A typo for print
D. A missing list item

REFERENCE LABEL: NOT PART OF THE QUESTION ID S89-F1-Q4
Artificial Intelligence, Set 89: Reading error messages · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.CODE.05 (local)

Learn
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Before asking a tool for a fix, what is a good first step?

A. Find the line and error type yourself
B. Restart the computer
C. Ignore the message
D. Delete the whole program and start again

REFERENCE LABEL: NOT PART OF THE QUESTION ID S89-F1-Q5
Artificial Intelligence, Set 89: Reading error messages · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.CODE.05 (local)

Learn
The line named in a traceback is where Python noticed the problem, not always where it started. A wrong value created earlier may only cause an error later. Fixes suggested by tools can hide errors instead of solving them, for example wrapping code so the error is silently ignored. Prefer fixes that address the cause, and after any fix, run the original failing case again to confirm the error is gone and the result is right.

An error appears on line 20, but the wrong value was created on line 5. Where is the cause?

A. Nowhere
B. Line 5
C. Line 1
D. Line 20

REFERENCE LABEL: NOT PART OF THE QUESTION ID S89-F2-Q1
Artificial Intelligence, Set 89: Reading error messages · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.CODE.05 (local)

Learn
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A tool "fixes" an error by making the program ignore all errors silently. What is wrong?

A. It uses Python
B. It is too short
C. Nothing
D. It hides the problem

REFERENCE LABEL: NOT PART OF THE QUESTION ID S89-F2-Q2
Artificial Intelligence, Set 89: Reading error messages · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.CODE.05 (local)

Learn
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After applying a fix, what should you do?

A. Rerun the original failing case

B. Delete the test

C. Ask for another fix

D. Move on without testing, since the tool already confirmed the fix would definitely work

REFERENCE LABEL: NOT PART OF THE QUESTION ID S89-F2-Q3
Artificial Intelligence, Set 89: Reading error messages · FORWARD 2: HANDLE THE COMPLICATIONS · Q3/5 · Outcome AI.CODE.05 (local)

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The error is gone, but the output is now wrong. Is it fixed?

A. Yes, if it is fast

B. Yes, because removing an error message is the same thing as fixing the problem

C. No; the result must also be right

D. It cannot be known

REFERENCE LABEL: NOT PART OF THE QUESTION ID S89-F2-Q4
Artificial Intelligence, Set 89: Reading error messages · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.CODE.05 (local)

Learn
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Which fix addresses the cause of IndexError from scores[3] on a 3-item list?

A. Hide the error message so that the program keeps running and prints nothing at all

B. Use a valid position, such as scores[2]

C. Rename the list

D. Add more print lines

REFERENCE LABEL: NOT PART OF THE QUESTION ID S89-F2-Q5
Artificial Intelligence, Set 89: Reading error messages · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.CODE.05 (local)

Set 90: Explaining and reviewing code

Outcome AI.CODE.06: Use explanations as a starting point and review code line by line against its requirements

Module: Coding: code assistance

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
Tools can EXPLAIN code in plain words, which is helpful for learning. Treat an explanation as a STARTING POINT: it may describe what code was meant to do rather than what it does. A CODE REVIEW checks code LINE BY LINE against its REQUIREMENTS. For each line ask: What does it do? Is it needed? Does it match the requirement? Is anything missing? Comments, names and explanations can all be wrong, so check the actual statements. Example: a comment says "return the largest number," but the code uses min(), which returns the smallest.

How should you treat a tool's explanation of code?

A. As useless
B. As a replacement for running it
C. As a starting point to check
D. As proof of what the code does

REFERENCE LABEL: NOT PART OF THE QUESTION ID S90-M-Q1
Artificial Intelligence, Set 90: Explaining and reviewing code · MAIN · Q1/5 · Outcome AI.CODE.06 (local)

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Prepared code example (Python 3). Shown only; nothing runs.

```
# Return the largest number in the list
def biggest(nums):
    return min(nums)

print(biggest([4, 9, 2]))
```

In this example, what does the program print?

A. 2

B. 4

C. 9, because the comment says the function returns the largest number in the list

D. 15

REFERENCE LABEL: NOT PART OF THE QUESTION ID S90-M-Q2
Artificial Intelligence, Set 90: Explaining and reviewing code · MAIN · Q2/5 · Outcome AI.CODE.06 (local)

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Prepared code example (Python 3). Shown only; nothing runs.

```
# Return the largest number in the list
def biggest(nums):
    return min(nums)

print(biggest([4, 9, 2]))
```

In the same example, what is wrong?

A. The comment uses the word "largest," which is not allowed in Python comments

B. Nothing

C. The list is too short

D. The code uses min() but the requirement is the largest

REFERENCE LABEL: NOT PART OF THE QUESTION ID S90-M-Q3
Artificial Intelligence, Set 90: Explaining and reviewing code · MAIN · Q3/5 · Outcome AI.CODE.06 (local)

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What does a code review check?

A. Only the file name
B. Each line against the requirements
C. Only the indentation
D. Only that the code has comments

REFERENCE LABEL: NOT PART OF THE QUESTION ID S90-M-Q4
Artificial Intelligence, Set 90: Explaining and reviewing code · MAIN · Q4/5 · Outcome AI.CODE.06 (local)

Learn
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What can be wrong in code besides the statements?

A. Comments, names and explanations
B. Only the spaces
C. Only the colors
D. Nothing; only the statements

REFERENCE LABEL: NOT PART OF THE QUESTION ID S90-M-Q5
Artificial Intelligence, Set 90: Explaining and reviewing code · MAIN · Q5/5 · Outcome AI.CODE.06 (local)

Learn
When you proofread a recipe for a friend, you do not just read the title ("Easy Chocolate Cake") and say "looks good." You check each line: is it "1 cup sugar" or "1 tablespoon"? Is the oven temperature there? Did they forget the eggs? The title is like a code comment: it tells you what it is supposed to be. The lines tell you what will actually happen. Code review is proofreading each line against what the code should do.

Why not stop at the recipe title?

A. Because cakes have no titles
B. It says what it should be, not what the lines do
C. Because titles are too short
D. Because titles of recipes are always wrong

REFERENCE LABEL: NOT PART OF THE QUESTION ID S90-B1-Q1
Artificial Intelligence, Set 90: Explaining and reviewing code · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.CODE.06 (local)

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The recipe title is like what in code?

A. An error message

B. A variable

C. A comment

D. A test that runs the code on several inputs and compares the results with answers

REFERENCE LABEL: NOT PART OF THE QUESTION ID S90-B1-Q2
Artificial Intelligence, Set 90: Explaining and reviewing code · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.CODE.06 (local)

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Checking "cup" versus "tablespoon" is like what?

A. Reading the title of the recipe twice to make sure that the cake is called the right thing

B. Choosing a font

C. Counting pages

D. Checking each line's details

REFERENCE LABEL: NOT PART OF THE QUESTION ID S90-B1-Q3
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Noticing missing eggs is like what in a code review?

A. Finding something missing
B. Renaming a variable
C. Running the code
D. Finding a misspelled comment

REFERENCE LABEL: NOT PART OF THE QUESTION ID S90-B1-Q4
Artificial Intelligence, Set 90: Explaining and reviewing code · BACKWARD 1 (concept simplification) · Q4/5 · Outcome AI.CODE.06 (local)

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What decides whether the cake will work?

A. The title of the recipe, which tells the baker exactly how the cake will turn out
B. The photo
C. The page color
D. The actual lines

REFERENCE LABEL: NOT PART OF THE QUESTION ID S90-B1-Q5
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Requirement: "Return the number of students who passed (score 60 or more)." The code counts scores above 60. What is the issue?

A. It uses numbers
B. It counts too many
C. It misses students who scored exactly 60
D. Nothing, since a score of exactly 60 is so rare

REFERENCE LABEL: NOT PART OF THE QUESTION ID S90-F1-Q1
Artificial Intelligence, Set 90: Explaining and reviewing code · FORWARD 1: APPLY IT · Q1/5 · Outcome AI.CODE.06 (local)

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A tool explains code as "sorts names alphabetically," but output shows them by length. What should you trust?

A. Neither
B. The output
C. The comment
D. The explanation

REFERENCE LABEL: NOT PART OF THE QUESTION ID S90-F1-Q2
Artificial Intelligence, Set 90: Explaining and reviewing code · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.CODE.06 (local)

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During review, you find a line whose purpose you cannot tell. What should you do?

A. Ask about it and test it before approving

B. Approve it anyway, since lines you do not understand are usually the most clever parts of code

C. Delete it without checking

D. Ignore it

REFERENCE LABEL: NOT PART OF THE QUESTION ID S90-F1-Q3
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Which review question asks about missing parts?

A. "Is the font readable?"

B. "Is it short?"

C. "Is the variable name long enough to describe exactly what the variable is used for?"

D. "Is anything the requirement needs not here?"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S90-F1-Q4
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Which is the best use of a code explanation?

A. To decide the requirement

B. To skip reading the code

C. To learn what to check, then verify by running

D. To replace testing entirely, since a clear explanation proves that the code is correct

REFERENCE LABEL: NOT PART OF THE QUESTION ID S90-F1-Q5
Artificial Intelligence, Set 90: Explaining and reviewing code · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.CODE.06 (local)

Learn
A good review also looks for what is NOT there: a missing check for empty input, a requirement that was skipped, or an unhandled error. Large changes are hard to review well; ask for small changes you can read in full. If you cannot understand a part of the code, that is a finding: ask for a simpler version or an explanation you can verify by running small examples, rather than approving it.

Code divides by the number of items but never checks for zero items. What kind of finding is this?

A. A comment problem
B. A missing check
C. No problem
D. A naming problem

REFERENCE LABEL: NOT PART OF THE QUESTION ID S90-F2-Q1
Artificial Intelligence, Set 90: Explaining and reviewing code · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.CODE.06 (local)

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A tool proposes changing 400 lines at once. What helps review?

A. Asking for smaller changes
B. Reading only the first line
C. Skipping review
D. Approving all 400 lines quickly

REFERENCE LABEL: NOT PART OF THE QUESTION ID S90-F2-Q2
Artificial Intelligence, Set 90: Explaining and reviewing code · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.CODE.06 (local)

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You cannot understand part of a generated function. What is the right response?

A. Pretend you understand

B. Remove the whole program

C. Ask for a simpler version or verify with small examples

D. Approve it, because code that is hard to understand is usually more efficient and better

REFERENCE LABEL: NOT PART OF THE QUESTION ID S90-F2-Q3
Artificial Intelligence, Set 90: Explaining and reviewing code · FORWARD 2: HANDLE THE COMPLICATIONS · Q3/5 · Outcome AI.CODE.06 (local)

Learn
A good review also looks for what is NOT there: a missing check for empty input, a requirement that was skipped, or an unhandled error. Large changes are hard to review well; ask for small changes you can read in full. If you cannot understand a part of the code, that is a finding: ask for a simpler version or an explanation you can verify by running small examples, rather than approving it.

A requirement says "log every failed login," but the code logs only some. What is this?

A. A faster design
B. A skipped requirement
C. A typo
D. A style choice

REFERENCE LABEL: NOT PART OF THE QUESTION ID S90-F2-Q4
Artificial Intelligence, Set 90: Explaining and reviewing code · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.CODE.06 (local)

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Which review finding is most serious?

A. A blank line
B. A long variable name
C. A comment that uses slightly informal language
D. An unhandled error that can crash the program

REFERENCE LABEL: NOT PART OF THE QUESTION ID S90-F2-Q5
Artificial Intelligence, Set 90: Explaining and reviewing code · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.CODE.06 (local)

Set 91: Version control: commits, differences and undoing

Outcome AI.CODE.07: Explain commits, differences (diffs) and reverting, and why to commit before accepting generated changes

Module: Coding: code assistance

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
VERSION CONTROL keeps a history of code changes. A popular tool for it is called Git. A COMMIT is a saved snapshot with a short message ("Fix average for empty list"). A DIFF (difference) shows what changed between versions: lines starting with "-" were removed and lines with "+" were added. REVERTING undoes a commit by adding a new commit with the opposite change; RESTORING returns files to how they were in a saved version. A good habit with AI help: COMMIT BEFORE accepting generated changes, so you can see exactly what the tool changed in the diff and undo it if it breaks something.

What is a commit?

A. A promise from the programmer to finish the project by a certain date in the future

B. A saved snapshot of code with a message

C. An error message

D. A deleted file

REFERENCE LABEL: NOT PART OF THE QUESTION ID S91-M-Q1
Artificial Intelligence, Set 91: Version control: commits, differences and undoing · MAIN · Q1/5 · Outcome AI.CODE.07 (local)

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In a diff, what does a line starting with "+" mean?

A. It was added
B. It is a comment
C. It has an error
D. It was removed

REFERENCE LABEL: NOT PART OF THE QUESTION ID S91-M-Q2
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Prepared example (a diff in a fictional project). Shown only; nothing runs.

```
- total = price * count
+ total = price + count
```

In this example, what changed?

A. The variable was renamed
B. Nothing changed
C. Multiplication became addition
D. Addition became multiplication

REFERENCE LABEL: NOT PART OF THE QUESTION ID S91-M-Q3
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What does reverting do?

A. Copies the project
B. Runs the tests
C. Deletes the whole project and its history
D. Undoes a commit with the opposite change

REFERENCE LABEL: NOT PART OF THE QUESTION ID S91-M-Q4
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Why commit before accepting generated changes?

A. To hide changes
B. To see what changed and undo it if needed
C. To make the code run faster
D. Because tools refuse to change code

REFERENCE LABEL: NOT PART OF THE QUESTION ID S91-M-Q5
Artificial Intelligence, Set 91: Version control: commits, differences and undoing · MAIN · Q5/5 · Outcome AI.CODE.07 (local)

Learn
In many video games you can SAVE before a hard level. If things go badly, you LOAD the save and try again. Each save is like a commit: a snapshot you can return to. Comparing two saves to see what changed, such as items gained or lost, is like a diff. Loading an earlier save is like restoring files from a commit. Saving before letting someone else play your game is like committing before accepting AI changes.

A game save is like what?

A. A test
B. A bug
C. A commit
D. A diff

REFERENCE LABEL: NOT PART OF THE QUESTION ID S91-B1-Q1
Artificial Intelligence, Set 91: Version control: commits, differences and undoing · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.CODE.07 (local)

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Comparing two saves to see what changed is like what?

A. A diff
B. A new game
C. A revert
D. A commit message

REFERENCE LABEL: NOT PART OF THE QUESTION ID S91-B1-Q2
Artificial Intelligence, Set 91: Version control: commits, differences and undoing · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.CODE.07 (local)

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Loading an earlier save is like what?

A. Restoring files from an earlier commit

B. Buying a game

C. Committing a brand-new change that adds a new level to the game for every player

D. Deleting the game

REFERENCE LABEL: NOT PART OF THE QUESTION ID S91-B1-Q3
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Saving before a friend plays is like what?

A. Committing before accepting AI changes

B. Deleting every save file so that your friend can start the whole game from the beginning

C. Changing the controls

D. Turning off the game

REFERENCE LABEL: NOT PART OF THE QUESTION ID S91-B1-Q4
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Why save before a hard level?

A. Because saving makes the next level easier to beat on the very first try
B. To end the game
C. To return if things go badly
D. To change the music

REFERENCE LABEL: NOT PART OF THE QUESTION ID S91-B1-Q5
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Ravi committed his working project, then let a tool edit it. Tests now fail, and he has not committed the edit. What can he do?

A. Nothing, because a tool's edits can never be undone
B. Delete the project
C. Restore the files from his last commit
D. Ignore the tests

REFERENCE LABEL: NOT PART OF THE QUESTION ID S91-F1-Q1
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A diff shows "- if age >= 18:" and "+ if age > 18:". What changed?

A. Nothing important

B. The rule now excludes exactly 18

C. The variable name

D. The rule now includes more ages

REFERENCE LABEL: NOT PART OF THE QUESTION ID S91-F1-Q2
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Which commit message is most useful?

A. "asdf"

B. "stuff"

C. "Fix average to handle an empty list"

D. "Changes," which tells everyone that something in the project was changed at that time

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Why read the diff after a tool edits code?

A. To rename files

B. To see exactly what it changed

C. Because diffs fix every bug automatically when they are opened and read by a person

D. To make the code longer

REFERENCE LABEL: NOT PART OF THE QUESTION ID S91-F1-Q4
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A diff shows only "+" lines and no "-" lines. What happened?

A. Lines were only added
B. Nothing changed
C. Every line in the file was deleted and replaced with completely different lines
D. Only comments were removed

REFERENCE LABEL: NOT PART OF THE QUESTION ID S91-F1-Q5
Artificial Intelligence, Set 91: Version control: commits, differences and undoing · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.CODE.07 (local)

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Commit messages should say what changed and why, so others (and future you) can understand the history. Small commits that each do one thing are easier to review and undo than one huge commit. Generated changes sometimes touch files you did not expect, such as settings or unrelated code; the diff reveals this. Never commit secrets like passwords, because version history keeps them even after you delete them from the latest version.

A diff from a tool edit shows changes to a settings file you did not ask about. What should you do?

A. Ignore the diff

B. Delete the settings file

C. Accept everything, since tools only change files that were needed for the requested task

D. Review those changes before accepting

REFERENCE LABEL: NOT PART OF THE QUESTION ID S91-F2-Q1
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Why prefer small commits?

A. Because large commits are not allowed by any version control tool that exists today

B. Because they hide changes

C. They are easier to review and undo

D. Because small commits use more storage

REFERENCE LABEL: NOT PART OF THE QUESTION ID S91-F2-Q2
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A password was committed and then deleted in the next commit. Is it gone?

A. No; history still has it
B. Yes, after one day
C. Only if the file is small
D. Yes

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What should a commit message say?

A. Nothing
B. What changed and why
C. Only the programmer's name
D. Only the date and time

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Which history is easiest to understand later?

A. No commits
B. Commits with random messages
C. One huge commit called "update"
D. Small commits with clear messages

REFERENCE LABEL: NOT PART OF THE QUESTION ID S91-F2-Q5
Artificial Intelligence, Set 91: Version control: commits, differences and undoing · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.CODE.07 (local)

Set 92: Dependencies and invented packages

Outcome AI.CODE.08: Check that a suggested library exists, is the intended one and is trustworthy before installing it

Module: Coding: code assistance

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

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Programs often use LIBRARIES, also called PACKAGES or DEPENDENCIES: code written by others that you install from a public collection, such as PyPI for Python. AI tools sometimes suggest a package that DOES NOT EXIST, and attackers can register those invented names with harmful code. This is sometimes called "slopsquatting." Researchers studying code-generating models (USENIX Security 2025) documented such invented package names. Before installing a suggested package, check that it exists, that it is the one you intended (not a look-alike name), and that it shows signs of trust: an active project, real documentation, and use by others.

What is a package or dependency?

A. A box that a computer is shipped in, which contains the cables and the instructions

B. Code written by others that your program installs and uses

C. An error message

D. A comment

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What can go wrong when a tool suggests a package?

A. It may not exist, and attackers can register the name

B. Nothing, because tools only ever suggest packages that have been checked for safety

C. It may have documentation

D. It may be too popular

REFERENCE LABEL: NOT PART OF THE QUESTION ID S92-M-Q2
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What is "slopsquatting"?

A. Writing very messy code on purpose so that other people find it hard to read and copy

B. Sorting a list

C. Deleting old packages

D. Registering package names that AI tools invent

REFERENCE LABEL: NOT PART OF THE QUESTION ID S92-M-Q3
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Which is a sign of a trustworthy package?

A. A name that looks like a famous one

B. No description

C. An active project with real documentation and users

D. A brand-new name that nobody has used, with no documentation

REFERENCE LABEL: NOT PART OF THE QUESTION ID S92-M-Q4
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Before installing a suggested package, what should you check first?

A. Its logo color
B. Nothing
C. That its name is short enough to type quickly
D. That it exists and is the intended one

REFERENCE LABEL: NOT PART OF THE QUESTION ID S92-M-Q5
Artificial Intelligence, Set 92: Dependencies and invented packages · MAIN · Q5/5 · Outcome AI.CODE.08 (local)

Learn
Before letting a contractor into your home to fix wiring, you check they are real: a business that exists, reviews from other customers, a license. You would be wary of someone with a name almost like a famous company's but one letter off, or a company nobody has heard of that showed up uninvited. Installing a package lets someone else's code into your program, so it deserves the same check.

Why check that a contractor exists?

A. Because contractors who exist always do perfect work, so existence is all that matters

B. To learn their age

C. Because you are letting them into your home

D. To get a discount

REFERENCE LABEL: NOT PART OF THE QUESTION ID S92-B1-Q1
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A name one letter off from a famous company is like what?

A. A look-alike package name

B. A test

C. A commit

D. A well-known package used by thousands of programmers and checked carefully

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Reviews from other customers are like what for a package?

A. Its name length
B. Signs it is used by others
C. Its error messages
D. The package's file size

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Installing a package is like what?

A. Letting someone else's work into your project
B. Deleting a file
C. Running a test
D. Writing every line of the program yourself

REFERENCE LABEL: NOT PART OF THE QUESTION ID S92-B1-Q4
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What is the lesson?

A. Install anything suggested
B. Only use paid packages
C. Never use code written by anyone else
D. Check a package before installing it

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A tool suggests "pip install easy-pdf-magic-tools." You find no project page. What should you do?

A. Rename it and install

B. Do not install it; look for a well-known alternative

C. Install it twice

D. Install it quickly, since a missing project page just means the package is very new

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A suggestion says "reqeusts." What should you notice?

A. It is a newer version

B. Nothing, because small spelling differences in package names do not matter at all

C. It is a look-alike of "requests"

D. It is a test file

REFERENCE LABEL: NOT PART OF THE QUESTION ID S92-F1-Q2
Artificial Intelligence, Set 92: Dependencies and invented packages · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.CODE.08 (local)

Learn
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Which package choice is safest for a class project?

A. A random new package

B. A package that a tool invented a name for, since invented packages are designed for the task

C. One with no documentation

D. One the course already uses

REFERENCE LABEL: NOT PART OF THE QUESTION ID S92-F1-Q3
Artificial Intelligence, Set 92: Dependencies and invented packages · FORWARD 1: APPLY IT · Q3/5 · Outcome AI.CODE.08 (local)

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A tool suggests a package to add two numbers. What is the better choice?

A. Write the one line yourself
B. Skip adding numbers
C. Install three packages
D. Install the package

REFERENCE LABEL: NOT PART OF THE QUESTION ID S92-F1-Q4
Artificial Intelligence, Set 92: Dependencies and invented packages · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.CODE.08 (local)

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Where should you check that a Python package exists?

A. In the tool's answer itself, since the tool would not suggest a package that did not exist

B. Its page on the official package index (PyPI)

C. Your memory

D. A random forum

REFERENCE LABEL: NOT PART OF THE QUESTION ID S92-F1-Q5
Artificial Intelligence, Set 92: Dependencies and invented packages · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.CODE.08 (local)

Learn
Look-alike names are a related risk: "reqeusts" instead of "requests," or a package with an extra word. Even real packages can be abandoned or compromised. Prefer well-known packages your course, school or workplace already uses, record exact versions, and do not add a dependency for something a few lines of your own code can do. When unsure, ask someone experienced before installing.

A real, popular package has had no updates for years and has open security warnings. What is the concern?

A. It has too many users
B. It is too popular
C. It may be abandoned
D. No concern

REFERENCE LABEL: NOT PART OF THE QUESTION ID S92-F2-Q1
Artificial Intelligence, Set 92: Dependencies and invented packages · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.CODE.08 (local)

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Why record exact package versions?

A. To make files longer
B. To hide which packages are used
C. Because package managers refuse to install anything
D. So builds are repeatable and changes are deliberate

REFERENCE LABEL: NOT PART OF THE QUESTION ID S92-F2-Q2
Artificial Intelligence, Set 92: Dependencies and invented packages · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.CODE.08 (local)

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A student is unsure whether a package is trustworthy. What is the best step?

A. Install it and hope
B. Ask someone experienced before installing
C. Share it with classmates
D. Install it on the school's shared computer first

REFERENCE LABEL: NOT PART OF THE QUESTION ID S92-F2-Q3
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Which name is the most suspicious?

A. "numpy-official-secure-v2"

B. "requests"

C. "numpy," which is a widely used package for working with numbers in Python programs

D. "pandas"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S92-F2-Q4
Artificial Intelligence, Set 92: Dependencies and invented packages · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.CODE.08 (local)

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Do real packages ever become unsafe?

A. Only if renamed

B. No; once a package is real and popular, it can never become unsafe in any way at all

C. Yes; they can be compromised

D. Only on weekends

REFERENCE LABEL: NOT PART OF THE QUESTION ID S92-F2-Q5
Artificial Intelligence, Set 92: Dependencies and invented packages · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.CODE.08 (local)

Set 93: Secrets and security in code

Outcome AI.CODE.09: Spot exposed secrets and unsafe handling of input in short code examples

Module: Coding: code assistance

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
A SECRET is a password, API key (a code that lets a program use a paid or private service) or similar credential. Never write secrets directly in code that will be shared or committed; store them separately, for example in a settings file kept private or an ENVIRONMENT VARIABLE (a value set outside the code). Also never paste secrets into a chat with an AI tool. A second risk is UNSAFE INPUT: code that takes what a user types and builds a command from it can be tricked. SQL (Structured Query Language) for databases is a classic case: inserting typed text straight into a query lets attackers change the query.

What is an API key?

A. An error message
B. A variable name
C. A code that lets a program use a private or paid service
D. A keyboard key that programmers press to run their code faster

REFERENCE LABEL: NOT PART OF THE QUESTION ID S93-M-Q1
Artificial Intelligence, Set 93: Secrets and security in code · MAIN · Q1/5 · Outcome AI.CODE.09 (local)

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Prepared code example (Python 3) with a fake key. Shown only; nothing runs.

```
API_KEY = "sk-live-4f9a2c7e1b"
print("Connecting with key", API_KEY)
```

In this example, what is the security problem?

A. The variable is called API_KEY in capital letters, which Python does not allow
B. Nothing; keys belong in code
C. The code prints a message
D. The key is written directly in the code

REFERENCE LABEL: NOT PART OF THE QUESTION ID S93-M-Q2
Artificial Intelligence, Set 93: Secrets and security in code · MAIN · Q2/5 · Outcome AI.CODE.09 (local)

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Where should secrets be stored?

A. In a comment at the top of the code, where every programmer can find them quickly

B. Separately, such as in an environment variable

C. In a chat with an AI tool

D. In the file name

REFERENCE LABEL: NOT PART OF THE QUESTION ID S93-M-Q3
Artificial Intelligence, Set 93: Secrets and security in code · MAIN · Q3/5 · Outcome AI.CODE.09 (local)

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Prepared code fragment (Python 3). Shown only; nothing runs.

```
name = input("Name: ")
query = "SELECT * FROM students WHERE name = '" + name + "'"
```

In this example, why is the query unsafe?

A. Typed text goes straight into the query
B. The table is called students
C. It uses the input function
D. The query is written in capital letters

REFERENCE LABEL: NOT PART OF THE QUESTION ID S93-M-Q4
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Should you paste a real API key into an AI chat to get help?

A. Yes, if it is short

B. Only on weekends

C. Yes, because chats with AI tools are completely private and are never stored anywhere

D. No; never paste secrets into a chat

REFERENCE LABEL: NOT PART OF THE QUESTION ID S93-M-Q5
Artificial Intelligence, Set 93: Secrets and security in code · MAIN · Q5/5 · Outcome AI.CODE.09 (local)

Learn
Leaving a house key under the doormat is convenient, but anyone who looks there can get in. Writing a password or API key directly in code is similar: anyone who sees the code, or a copy of it, can use the key. Keeping the key in your pocket, separate from the house, is like storing secrets outside the code. And if the key might have been copied, you change the lock rather than just moving the key.

Why is a key under the doormat risky?

A. Because keys are heavy

B. Because doormats damage keys over time, making them stop working in the lock

C. Anyone who looks can find it

D. Because doors are locked

REFERENCE LABEL: NOT PART OF THE QUESTION ID S93-B1-Q1
Artificial Intelligence, Set 93: Secrets and security in code · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.CODE.09 (local)

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A secret written in code is like what?

A. A locked safe
B. A key under the doormat
C. A doorbell
D. A key kept safely

REFERENCE LABEL: NOT PART OF THE QUESTION ID S93-B1-Q2
Artificial Intelligence, Set 93: Secrets and security in code · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.CODE.09 (local)

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Keeping the key in your pocket is like what?

A. Storing secrets outside the code
B. Giving it away
C. Posting it online
D. Writing the secret into a comment

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If the key may have been copied, what should you do?

A. Ignore it
B. Paint the door
C. Change the lock
D. Move the key

REFERENCE LABEL: NOT PART OF THE QUESTION ID S93-B1-Q4
Artificial Intelligence, Set 93: Secrets and security in code · BACKWARD 1 (concept simplification) · Q4/5 · Outcome AI.CODE.09 (local)

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In code, "changing the lock" means what?

A. Replacing the exposed secret with a new one
B. Adding a comment
C. Renaming the variable
D. Deleting the line with the secret from the newest version of the code only

REFERENCE LABEL: NOT PART OF THE QUESTION ID S93-B1-Q5
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Mia shares her project online and realizes her real API key is in the code. What should she do first?

A. Ignore it

B. Delete the key from the code and assume the problem is solved, since nobody saw it yet

C. Make the repository bigger

D. Change the key at the service right away

REFERENCE LABEL: NOT PART OF THE QUESTION ID S93-F1-Q1
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A tool asks you to paste your password so it can "test the login code." What should you do?

A. Paste it, because tools need the real password to check whether the login code works

B. Refuse; use a fake test password

C. Paste half of it

D. Email it instead

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Which line handles a secret more safely?

A. Printing the key at startup
B. Saving the key in the file name
C. Reading the key from an environment variable
D. Writing the key in a comment next to the code

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A form's code builds a database query by adding typed text. What is the risk?

A. Nothing
B. The text will be misspelled
C. The query will run more slowly
D. An attacker can change the query

REFERENCE LABEL: NOT PART OF THE QUESTION ID S93-F1-Q4
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Generated code contains "API_KEY = 'your-key-here'." What is the safe next step?

A. Load the real key from a private setting
B. Delete the line
C. Share the placeholder
D. Replace the placeholder text with your real key

REFERENCE LABEL: NOT PART OF THE QUESTION ID S93-F1-Q5
Artificial Intelligence, Set 93: Secrets and security in code · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.CODE.09 (local)

Learn
Generated code often uses placeholder secrets ("API_KEY = 'your-key-here'"), and people replace them with real keys in the same spot, then share the file. If a secret is exposed, CHANGE IT (rotate it) at the service right away; deleting it from the code is not enough. For input, use the safe methods a language provides, such as PARAMETERIZED QUERIES that keep user text separate from the command, and treat everything a user types as untrusted.

Why is deleting an exposed key from the code not enough?

A. Because keys grow back
B. It may already have been copied
C. Because code cannot be edited
D. Because deleting lines of code

REFERENCE LABEL: NOT PART OF THE QUESTION ID S93-F2-Q1
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What does "rotate a secret" mean?

A. Write it backward
B. Encrypt the file name
C. Move the secret to a different line
D. Replace it with a new one at the service

REFERENCE LABEL: NOT PART OF THE QUESTION ID S93-F2-Q2
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What do parameterized queries do?

A. Keep user text separate from the command

B. Print the query

C. Remove the database

D. Make the database query run on a schedule every hour without any user input at all

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How should code treat what a user types?

A. As a secret
B. As untrusted
C. As a comment
D. As fully trusted

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A tutorial's generated code prints the API key at startup "for debugging." What is the risk?

A. The program runs slower

B. No risk, since printed text disappears from the screen as soon as the program finishes

C. The key can appear in logs or screenshots

D. The key gets longer

REFERENCE LABEL: NOT PART OF THE QUESTION ID S93-F2-Q5
Artificial Intelligence, Set 93: Secrets and security in code · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.CODE.09 (local)

Set 94: Maintaining code you did not write

Outcome AI.CODE.10: Keep generated code understandable, documented and owned by someone who can maintain it

Module: Coding: code assistance

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
Code is usually read and changed far more often than it is written. Generated code you cannot explain becomes a problem when it breaks or needs a change. Keep code MAINTAINABLE: clear NAMES (total_price, not x2), short FUNCTIONS that each do one thing, COMMENTS that explain why, and a README (a document explaining what the project does and how to run it). Every piece of code in use should have an OWNER: a person who understands it and is responsible for fixing it. If no one can explain how it works, it is not ready to rely on.

Is code read or written more often?

A. Read and changed more often

B. Neither

C. Written more often, because each line of code is written many times and read only once

D. Only written

REFERENCE LABEL: NOT PART OF THE QUESTION ID S94-M-Q1
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Prepared code example (Python 3). Shown only; nothing runs.

```
def f(a, b):
    return a * b * 1.0825

def price_with_tax(subtotal, quantity):
    TAX_RATE = 0.0825  # local sales tax
    return subtotal * quantity * (1 + TAX_RATE)

print(round(f(10, 2), 2), round(price_with_tax(10, 2), 2))
```

In this example, which function is easier to maintain?

A. Neither

B. price_with_tax

C. f, because short names like f, a and b are faster to type and always easier to understand

D. Both equally

REFERENCE LABEL: NOT PART OF THE QUESTION ID S94-M-Q2
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def f(a, b):
    return a * b * 1.0825

def price_with_tax(subtotal, quantity):
    TAX_RATE = 0.0825  # local sales tax
    return subtotal * quantity * (1 + TAX_RATE)

print(round(f(10, 2), 2), round(price_with_tax(10, 2), 2))
```

In the same example, do both functions give the same result?

A. No; f gives 20
B. No; one crashes
C. No, because functions with different names always calculate different results
D. Yes, 21.65

REFERENCE LABEL: NOT PART OF THE QUESTION ID S94-M-Q3
Artificial Intelligence, Set 94: Maintaining code you did not write · MAIN · Q3/5 · Outcome AI.CODE.10 (local)

Learn
Code is usually read and changed far more often than it is written. Generated code you cannot explain becomes a problem when it breaks or needs a change. Keep code MAINTAINABLE: clear NAMES (total_price, not x2), short FUNCTIONS that each do one thing, COMMENTS that explain why, and a README (a document explaining what the project does and how to run it). Every piece of code in use should have an OWNER: a person who understands it and is responsible for fixing it. If no one can explain how it works, it is not ready to rely on.

What is a README?

A. A password file
B. An error log
C. A document explaining what a project does and how to run it
D. A file that the computer reads before every program

REFERENCE LABEL: NOT PART OF THE QUESTION ID S94-M-Q4
Artificial Intelligence, Set 94: Maintaining code you did not write · MAIN · Q4/5 · Outcome AI.CODE.10 (local)

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Who should own a piece of code in use?

A. Nobody
B. The AI tool that wrote it, which will maintain and fix it forever without any help
C. Whoever used it last
D. A person who understands it and fixes it

REFERENCE LABEL: NOT PART OF THE QUESTION ID S94-M-Q5
Artificial Intelligence, Set 94: Maintaining code you did not write · MAIN · Q5/5 · Outcome AI.CODE.10 (local)

Learn
In a shared kitchen with LABELED jars ("sugar," "salt"), anyone can cook. In a kitchen of unlabeled jars, only the person who filled them knows which is which, and when they are away, someone puts salt in the cake. Code works the same way: clear names and comments are labels. A README is the note on the fridge explaining how things work. The OWNER is the person who restocks and keeps it organized.

Labeled jars are like what in code?

A. Clear names and comments
B. Error messages
C. Passwords
D. Very short names like x and y

REFERENCE LABEL: NOT PART OF THE QUESTION ID S94-B1-Q1
Artificial Intelligence, Set 94: Maintaining code you did not write · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.CODE.10 (local)

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Unlabeled jars cause what problem?

A. Food is cheaper

B. Only one person knows what is what

C. Everyone in the kitchen always knows exactly what is in every jar without looking

D. Jars break

REFERENCE LABEL: NOT PART OF THE QUESTION ID S94-B1-Q2
Artificial Intelligence, Set 94: Maintaining code you did not write · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.CODE.10 (local)

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The note on the fridge is like what?

A. A commit message that explains a single change made to one line of the code
B. A bug
C. A README
D. A test

REFERENCE LABEL: NOT PART OF THE QUESTION ID S94-B1-Q3
Artificial Intelligence, Set 94: Maintaining code you did not write · BACKWARD 1 (concept simplification) · Q3/5 · Outcome AI.CODE.10 (local)

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The person who keeps the kitchen organized is like what?

A. The fridge
B. The code's owner
C. A guest
D. The AI tool

REFERENCE LABEL: NOT PART OF THE QUESTION ID S94-B1-Q4
Artificial Intelligence, Set 94: Maintaining code you did not write · BACKWARD 1 (concept simplification) · Q4/5 · Outcome AI.CODE.10 (local)

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Salt in the cake is like what in code?

A. A successful new feature that customers asked for and that works exactly as planned

B. A faster program

C. A mistake caused by unclear code

D. A new commit

REFERENCE LABEL: NOT PART OF THE QUESTION ID S94-B1-Q5
Artificial Intelligence, Set 94: Maintaining code you did not write · BACKWARD 1 (concept simplification) · Q5/5 · Outcome AI.CODE.10 (local)

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A club uses a generated script to track dues, but nobody can explain how it works. What is the risk?

A. No one can fix it when it breaks

B. It uses too little memory

C. No risk, because generated scripts never break and never need to be changed by anyone

D. It will run too fast

REFERENCE LABEL: NOT PART OF THE QUESTION ID S94-F1-Q1
Artificial Intelligence, Set 94: Maintaining code you did not write · FORWARD 1: APPLY IT · Q1/5 · Outcome AI.CODE.10 (local)

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Which variable name is clearest?

A. thing
B. tp
C. x2
D. total_price

REFERENCE LABEL: NOT PART OF THE QUESTION ID S94-F1-Q2
Artificial Intelligence, Set 94: Maintaining code you did not write · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.CODE.10 (local)

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A generated function does five different jobs. What improves it?

A. Renaming it to "do_all"
B. Removing all comments
C. Splitting it into smaller functions
D. Adding a sixth job to the same function

REFERENCE LABEL: NOT PART OF THE QUESTION ID S94-F1-Q3
Artificial Intelligence, Set 94: Maintaining code you did not write · FORWARD 1: APPLY IT · Q3/5 · Outcome AI.CODE.10 (local)

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Which comment is most useful?

A. "# This line of code does something," which reminds readers that the line is important

B. "# code"

C. "# hi"

D. "# Round down because the store never charges partial cents"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S94-F1-Q4
Artificial Intelligence, Set 94: Maintaining code you did not write · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.CODE.10 (local)

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Before relying on generated code at work, what should be true?

A. It should have no comments
B. Someone can explain how it works
C. It should be secret
D. It should be long

REFERENCE LABEL: NOT PART OF THE QUESTION ID S94-F1-Q5
Artificial Intelligence, Set 94: Maintaining code you did not write · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.CODE.10 (local)

Learn
Maintenance problems build up quietly. "Magic numbers" such as 0.0825 with no explanation leave the next person guessing (is it a tax rate?). Duplicated code means a fix in one place is missed in another. When a tool generates a large solution, ask for a simpler version, add tests that describe how it should behave, and write down any decisions. When the owner leaves a team or class, hand over the knowledge, not just the files.

Code contains 0.0825 with no explanation. What is this called?

A. A magic number
B. A test
C. A comment
D. A secret key

REFERENCE LABEL: NOT PART OF THE QUESTION ID S94-F2-Q1
Artificial Intelligence, Set 94: Maintaining code you did not write · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.CODE.10 (local)

Learn
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The same calculation is copied in three places. What problem follows?

A. It is easier to read

B. A fix in one place may be missed in others

C. The program runs faster

D. Nothing, because copied code always gets updated in every place at the same time

REFERENCE LABEL: NOT PART OF THE QUESTION ID S94-F2-Q2
Artificial Intelligence, Set 94: Maintaining code you did not write · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.CODE.10 (local)

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A tool generates a 300-line solution for a simple task. What should you ask for?

A. A simpler version
B. An even longer version with more features, so it is ready for any future need
C. Nothing; use it
D. A version without tests

REFERENCE LABEL: NOT PART OF THE QUESTION ID S94-F2-Q3
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The student who built the class website is graduating. What should happen?

A. Change the password and tell no one
B. Delete the website
C. Hand over knowledge, not just files
D. Nothing

REFERENCE LABEL: NOT PART OF THE QUESTION ID S94-F2-Q4
Artificial Intelligence, Set 94: Maintaining code you did not write · FORWARD 2: HANDLE THE COMPLICATIONS · Q4/5 · Outcome AI.CODE.10 (local)

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Why add tests to generated code you will keep?

A. To slow it down

B. To hide bugs

C. Because tests make the code shorter and remove the need for any comments or README

D. They describe and protect how it should behave

REFERENCE LABEL: NOT PART OF THE QUESTION ID S94-F2-Q5
Artificial Intelligence, Set 94: Maintaining code you did not write · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.CODE.10 (local)

Set 95: Chatbot, tool-using workflow or agent

Outcome AI.AGENT.01: Tell a chatbot, a fixed tool-using workflow and an agent that chooses its own steps

Module: Coding: agents and automation

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
Three setups are often all called "AI," but they differ in who decides the steps. A CHATBOT answers in a conversation; you read the reply and decide what to do. A TOOL-USING WORKFLOW follows FIXED steps that a person designed, using a model for some steps, for example "when a form arrives, summarize it, then email the summary to the manager." An AGENT is given a GOAL and TOOLS (such as search, email or files) and CHOOSES its own next steps, checking results as it goes. More freedom to choose means more ways to go wrong, so agents need the most limits and oversight.

What does a chatbot do?

A. Prints documents

B. Follows a fixed schedule

C. Answers in a conversation while you decide what to do

D. Takes actions in the world on its own, such as sending emails

REFERENCE LABEL: NOT PART OF THE QUESTION ID S95-M-Q1
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What makes a workflow "fixed"?

A. It never uses a model
B. It has no steps
C. The model chooses each new step by itself
D. A person designed its steps in advance

REFERENCE LABEL: NOT PART OF THE QUESTION ID S95-M-Q2
Artificial Intelligence, Set 95: Chatbot, tool-using workflow or agent · MAIN · Q2/5 · Outcome AI.AGENT.01 (local)

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What makes something an agent?

A. It has a friendly name
B. It chooses its own next steps toward a goal
C. It runs on a timer
D. It answers one question at a time and waits

REFERENCE LABEL: NOT PART OF THE QUESTION ID S95-M-Q3
Artificial Intelligence, Set 95: Chatbot, tool-using workflow or agent · MAIN · Q3/5 · Outcome AI.AGENT.01 (local)

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"When a form arrives, summarize it, then email the manager" describes what?

A. A tool-using workflow
B. A chatbot
C. A search engine
D. An agent

REFERENCE LABEL: NOT PART OF THE QUESTION ID S95-M-Q4
Artificial Intelligence, Set 95: Chatbot, tool-using workflow or agent · MAIN · Q4/5 · Outcome AI.AGENT.01 (local)

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Which needs the most limits and oversight?

A. A calculator
B. A fixed workflow
C. A chatbot
D. An agent

REFERENCE LABEL: NOT PART OF THE QUESTION ID S95-M-Q5
Artificial Intelligence, Set 95: Chatbot, tool-using workflow or agent · MAIN · Q5/5 · Outcome AI.AGENT.01 (local)

Learn
A GPS that you ASK for directions is like a chatbot: it suggests, and you drive. A delivery company's FIXED ROUTE, same stops in the same order every day, is like a workflow: designed in advance. A PERSONAL ASSISTANT told "get me to the airport by 6" who chooses taxis, trains and timing is like an agent: given a goal, choosing steps. The assistant is most flexible, and also the one whose choices you most need to check.

A GPS you ask for directions is like what?

A. A calendar
B. A chatbot
C. A workflow
D. An agent

REFERENCE LABEL: NOT PART OF THE QUESTION ID S95-B1-Q1
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A fixed delivery route is like what?

A. A map
B. A chatbot
C. A workflow
D. An agent

REFERENCE LABEL: NOT PART OF THE QUESTION ID S95-B1-Q2
Artificial Intelligence, Set 95: Chatbot, tool-using workflow or agent · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.AGENT.01 (local)

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An assistant choosing taxis and trains is like what?

A. An agent
B. A timetable
C. A fixed workflow
D. A chatbot

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Which option is most flexible?

A. All are equal
B. The personal assistant
C. The GPS
D. The fixed delivery route

REFERENCE LABEL: NOT PART OF THE QUESTION ID S95-B1-Q4
Artificial Intelligence, Set 95: Chatbot, tool-using workflow or agent · BACKWARD 1 (concept simplification) · Q4/5 · Outcome AI.AGENT.01 (local)

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Whose choices most need checking?

A. The assistant's
B. The route's
C. No one's
D. The GPS's

REFERENCE LABEL: NOT PART OF THE QUESTION ID S95-B1-Q5
Artificial Intelligence, Set 95: Chatbot, tool-using workflow or agent · BACKWARD 1 (concept simplification) · Q5/5 · Outcome AI.AGENT.01 (local)

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Lena types questions to a study helper and decides what to write. What is it?

A. A fixed workflow
B. A spreadsheet
C. A chatbot
D. An agent

REFERENCE LABEL: NOT PART OF THE QUESTION ID S95-F1-Q1
Artificial Intelligence, Set 95: Chatbot, tool-using workflow or agent · FORWARD 1: APPLY IT · Q1/5 · Outcome AI.AGENT.01 (local)

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Every night, a system collects new orders, uses a model to tag them, and files them. What is it?

A. A chatbot
B. A calculator
C. An agent
D. A fixed workflow

REFERENCE LABEL: NOT PART OF THE QUESTION ID S95-F1-Q2
Artificial Intelligence, Set 95: Chatbot, tool-using workflow or agent · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.AGENT.01 (local)

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Told "book the cheapest flight to Denver next week," a system decides which searches to run, changes its plan when a flight is sold out, and then books. What is it?

A. A search engine
B. A fixed workflow
C. An agent
D. A chatbot

REFERENCE LABEL: NOT PART OF THE QUESTION ID S95-F1-Q3
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Which system can affect the world most directly?

A. A printed manual

B. A calculator

C. A chatbot that only shows text answers on a screen for the person to read and decide

D. An agent that can send and buy

REFERENCE LABEL: NOT PART OF THE QUESTION ID S95-F1-Q4
Artificial Intelligence, Set 95: Chatbot, tool-using workflow or agent · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.AGENT.01 (local)

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A workflow uses a model only to summarize; a person designed every other step. What decides the order of steps?

A. The person's design

B. Chance

C. The customer

D. The model, which rearranges the order of the steps every time the workflow is started

REFERENCE LABEL: NOT PART OF THE QUESTION ID S95-F1-Q5
Artificial Intelligence, Set 95: Chatbot, tool-using workflow or agent · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.AGENT.01 (local)

Learn
Labels in product marketing are loose: something sold as an "agent" may be a fixed workflow, and a chatbot may quietly use tools such as search. Ask practical questions instead. Who decides the next step, a person, a fixed design or the model? What can it actually do in the world: only show text, or also send, buy, delete or post? Can it act without a person approving? The answers tell you how much checking and how many limits it needs.

A product is marketed as an "agent" but always runs the same three steps. What is it really?

A. A database
B. A fixed workflow
C. A chatbot
D. An agent

REFERENCE LABEL: NOT PART OF THE QUESTION ID S95-F2-Q1
Artificial Intelligence, Set 95: Chatbot, tool-using workflow or agent · FORWARD 2: HANDLE THE COMPLICATIONS · Q1/5 · Outcome AI.AGENT.01 (local)

Learn
Labels in product marketing are loose: something sold as an "agent" may be a fixed workflow, and a chatbot may quietly use tools such as search. Ask practical questions instead. Who decides the next step, a person, a fixed design or the model? What can it actually do in the world: only show text, or also send, buy, delete or post? Can it act without a person approving? The answers tell you how much checking and how many limits it needs.

Which question best reveals the risk of a system?

A. "Is it new?"
B. "Is it popular?"
C. "Does it have a friendly name and a nice logo?"
D. "Can it act without a person approving?"

REFERENCE LABEL: NOT PART OF THE QUESTION ID S95-F2-Q2
Artificial Intelligence, Set 95: Chatbot, tool-using workflow or agent · FORWARD 2: HANDLE THE COMPLICATIONS · Q2/5 · Outcome AI.AGENT.01 (local)

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A chatbot quietly searches the web before answering. Does that make it an agent?

A. Not necessarily; it depends on who chooses the next steps
B. No, chatbots cannot use tools
C. Yes, because any system that uses a tool is an agent
D. Yes, if it is fast

REFERENCE LABEL: NOT PART OF THE QUESTION ID S95-F2-Q3
Artificial Intelligence, Set 95: Chatbot, tool-using workflow or agent · FORWARD 2: HANDLE THE COMPLICATIONS · Q3/5 · Outcome AI.AGENT.01 (local)

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Which ability needs the most limits?

A. Answering a quiz question
B. Deleting files
C. Suggesting a title
D. Showing a short text summary

REFERENCE LABEL: NOT PART OF THE QUESTION ID S95-F2-Q4
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Who decides the next step tells you what?

A. Nothing important, since every AI system needs exactly the same amount of checking

B. Its speed

C. How much checking the system needs

D. Its price

REFERENCE LABEL: NOT PART OF THE QUESTION ID S95-F2-Q5
Artificial Intelligence, Set 95: Chatbot, tool-using workflow or agent · FORWARD 2: HANDLE THE COMPLICATIONS · Q5/5 · Outcome AI.AGENT.01 (local)

Set 96: Goals, tools, plans and stopping

Outcome AI.AGENT.02: Describe an agent's goal, tools, plan, feedback and stopping condition

Module: Coding: agents and automation

Course scope: general education for adults and high-school students, not legal, medical or financial advice. No AI tool, account, download or paid service is needed: every example is prepared in advance and shown as text, and nothing runs. Examples labeled "fictional" were written for this course and are not real outputs of any named product. Laws, products and prices change; dated lessons say what to check. People and organizations in scenarios are fictional unless a lesson names a real case.

Learn
An agent can be described with five parts. GOAL: what it is trying to achieve ("find three venues under 500 dollars for 40 people"). TOOLS: what it can use (search, calendar, email). PLAN: the steps it chooses (search, compare, list). FEEDBACK: the results it checks to decide what next (a venue is full, so look again). STOPPING CONDITION: when it must stop (three venues found, 20 searches done, or a question it cannot answer). A vague goal or missing stopping condition can make an agent wander, repeat steps, or take actions nobody intended.

"Find three venues under 500 dollars for 40 people" is which part?

A. Tool, because finding venues is something that a search tool does for the agent

B. Goal

C. Feedback

D. Stopping condition

REFERENCE LABEL: NOT PART OF THE QUESTION ID S96-M-Q1
Artificial Intelligence, Set 96: Goals, tools, plans and stopping · MAIN · Q1/5 · Outcome AI.AGENT.02 (local)

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Search, calendar and email are which part?

A. Plan
B. Feedback
C. Goals
D. Tools

REFERENCE LABEL: NOT PART OF THE QUESTION ID S96-M-Q2
Artificial Intelligence, Set 96: Goals, tools, plans and stopping · MAIN · Q2/5 · Outcome AI.AGENT.02 (local)

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"A venue is full, so look again" is which part?

A. Goal, because looking again for another venue is the main thing the agent wants to do

B. Tool

C. Feedback

D. Stopping condition

REFERENCE LABEL: NOT PART OF THE QUESTION ID S96-M-Q3
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"Stop after 20 searches" is which part?

A. Stopping condition

B. Goal

C. Tool

D. Plan, because twenty searches is the list of steps the agent chooses to take in order

REFERENCE LABEL: NOT PART OF THE QUESTION ID S96-M-Q4
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What can a missing stopping condition cause?

A. Fewer tools
B. A shorter plan
C. Wandering or repeating steps
D. Faster and better results

REFERENCE LABEL: NOT PART OF THE QUESTION ID S96-M-Q5
Artificial Intelligence, Set 96: Goals, tools, plans and stopping · MAIN · Q5/5 · Outcome AI.AGENT.02 (local)

Learn
Sending a friend shopping has the same parts. GOAL: "ingredients for tacos for 6." TOOLS: your list, money and a car. PLAN: which stores and aisles. FEEDBACK: "they are out of tortillas, so try the bakery." STOPPING CONDITION: "come back once you have everything or after spending 40 dollars." Without that last part, your friend might drive to five stores looking for one item.

"Ingredients for tacos for 6" is which part?

A. Feedback
B. Goal
C. Tool
D. Plan

REFERENCE LABEL: NOT PART OF THE QUESTION ID S96-B1-Q1
Artificial Intelligence, Set 96: Goals, tools, plans and stopping · BACKWARD 1 (concept simplification) · Q1/5 · Outcome AI.AGENT.02 (local)

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Money and a car are which part?

A. Stopping condition
B. Goals, because the friend is trying to get money and a car during the trip
C. Feedback
D. Tools

REFERENCE LABEL: NOT PART OF THE QUESTION ID S96-B1-Q2
Artificial Intelligence, Set 96: Goals, tools, plans and stopping · BACKWARD 1 (concept simplification) · Q2/5 · Outcome AI.AGENT.02 (local)

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"Out of tortillas, so try the bakery" is which part?

A. Feedback

B. Stopping condition, since running out of tortillas means the shopping is finished

C. Goal

D. Tool

REFERENCE LABEL: NOT PART OF THE QUESTION ID S96-B1-Q3
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"Come back after spending 40 dollars" is which part?

A. Tool

B. Feedback

C. Stopping condition

D. Goal, because spending forty dollars is the main reason for going shopping at all

REFERENCE LABEL: NOT PART OF THE QUESTION ID S96-B1-Q4
Artificial Intelligence, Set 96: Goals, tools, plans and stopping · BACKWARD 1 (concept simplification) · Q4/5 · Outcome AI.AGENT.02 (local)

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What happens without a stopping condition?

A. Your friend may keep searching too long
B. The tacos taste better
C. Nothing changes
D. Your friend will always come back sooner

REFERENCE LABEL: NOT PART OF THE QUESTION ID S96-B1-Q5
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An agent is told "organize my inbox" with no other details. What is missing?

A. More tools, because an agent with access to more tools always understands vague goals

B. A clear goal and stopping condition

C. Nothing

D. A shorter name

REFERENCE LABEL: NOT PART OF THE QUESTION ID S96-F1-Q1
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Which stopping condition is clearest?

A. "Stop sometime."
B. "Never stop."
C. "Stop whenever you feel that the inbox looks nice."
D. "Stop after sorting 200 emails or 10 minutes."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S96-F1-Q2
Artificial Intelligence, Set 96: Goals, tools, plans and stopping · FORWARD 1: APPLY IT · Q2/5 · Outcome AI.AGENT.02 (local)

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An agent searching for flights finds one sold out and searches again. What part is at work?

A. Feedback
B. Stopping condition
C. Goal, because a sold-out flight is a new goal that the agent must achieve next
D. Tool

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An agent can use a calendar but not email. Which part does this describe?

A. Goal
B. Feedback
C. Tools
D. Plan

REFERENCE LABEL: NOT PART OF THE QUESTION ID S96-F1-Q4
Artificial Intelligence, Set 96: Goals, tools, plans and stopping · FORWARD 1: APPLY IT · Q4/5 · Outcome AI.AGENT.02 (local)

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Before a high-stakes task, what is a sensible request?

A. "Act as fast as possible."

B. "Show me your plan before acting."

C. "Use every tool you can."

D. "Go ahead with whatever plan you think is best, and tell me about it next week."

REFERENCE LABEL: NOT PART OF THE QUESTION ID S96-F1-Q5
Artificial Intelligence, Set 96: Goals, tools, plans and stopping · FORWARD 1: APPLY IT · Q5/5 · Outcome AI.AGENT.02 (local)

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Goals can be misread. "Get the most sign-ups" might lead an agent to send far too many emails, because nothing said not to. Good goals include LIMITS ("no more than one email per person") and say what to do when uncertain ("ask me before contacting anyone new"). A stopping condition should include a budget of time, steps or money, so a confused agent stops instead of looping. Ask the agent to report its plan before acting when the stakes are high.

An agent told "get the most sign-ups" sends ten emails to each person. What went wrong?

A. The sign-up form was wrong
B. The emails were too short
C. The agent was broken
D. The goal had no limits

REFERENCE LABEL: NOT PART OF THE QUESTION ID S96-F2-Q1
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Which goal includes a useful limit?

A. "Invite as many people as you possibly can, using any method you think will work."

B. "Do your best."

C. "Invite club members, no more than one email each."

D. "Invite people."

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What should a stopping condition include?

A. A budget of time, steps or money

B. Only a friendly message

C. Nothing specific, since agents always know by themselves when it is the right time to stop

D. A longer goal

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An agent keeps repeating the same search without progress. What would have prevented this?

A. A faster computer
B. A longer goal
C. Giving the agent more tools to use
D. A step limit in its stopping condition

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Which instruction tells an agent what to do when unsure?

A. "Use your best judgment always."
B. "Ask me before contacting anyone new."
C. "Never ask questions."
D. "Guess whenever you are unsure."

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